ALKYL HALIDES

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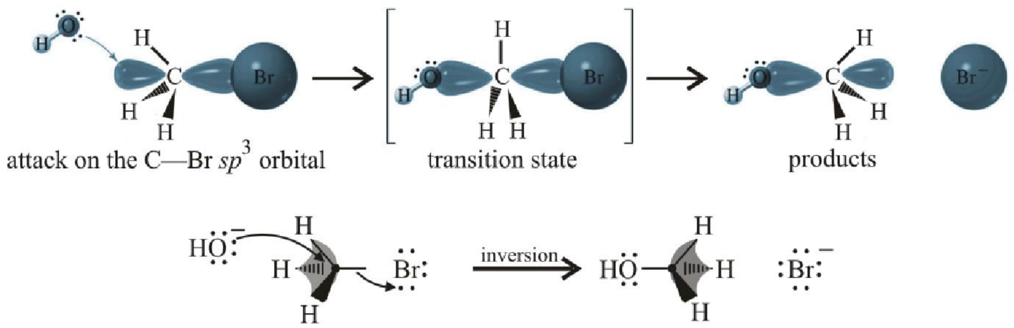
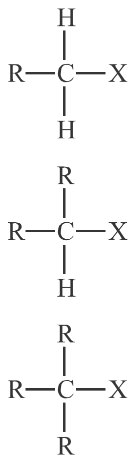


THEORY

1. INTRODUCTION

The **alkyl halides** or **haloalkanes** are a group of chemical compounds, derived from alkanes containing one or more halogens. They are used commonly as flame retardants, fire extinguishers, refrigerants, propellants, solvents and pharmaceuticals. The alkyl halides are classified broadly into three categories based on type of carbon atom to which the halogen atom is attached.



1° Primary

2° Secondary

3° Tertiary

X may be F, Cl, Br or I.

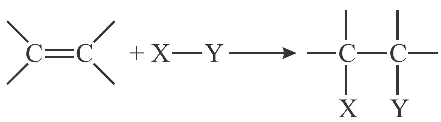
2. REACTIONS IN ORGANIC CHEMISTRY

There are various types of reactions possible in organic compounds depending on reaction conditions and attacking reagent. Reactions in organic chemistry are classified into three categories :

2.1 Addition Reaction

This reaction involves addition of groups to a π bond.

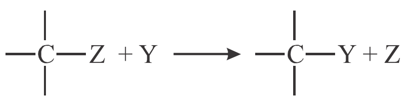
Example - 1



2.2 Substitution Reaction

This reaction involves the replacement of an atom or a group of atoms by another atom or group of atoms.

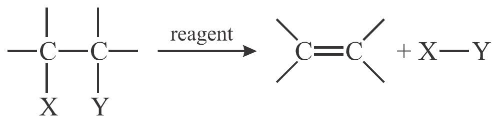
Example - 2



2.3 Elimination Reaction

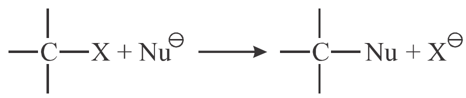
This reaction involves the loss of atoms or groups of atoms to form an unsaturated compound.

Example - 3



3. NUCLEOPHILIC SUBSTITUTIONS REACTIONS

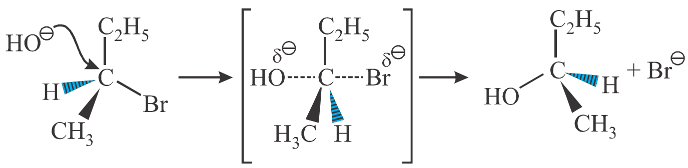
Example - 4



The replacement of halogen atom (leaving group) by the attacking nucleophile is called nucleophilic substitution reaction at sp^3 carbon. This reaction was studied in great detail and two extreme mechanisms have been outlined to explain the course of the reaction.

3.1 Substitution Nucleophilic Bimolecular – S_N2

Example - 5



Transition State

Note.. **Key Features of S_N2 Mechanism**

1. Single step reaction.
2. Rate = k [RX] [Nu]
3. No intermediate is formed. Reaction goes through a transition state.
4. Rearrangement is not observed.
5. Inversion of configuration is observed.
6. Order of reactivity of alkyl halides :

$$CH_3X > 1° > 2° > 3°$$

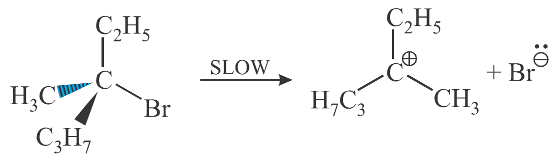
This can be attributed to the steric hinderance to back side attack of nucleophile.

7. Favoured by aprotic solvents.

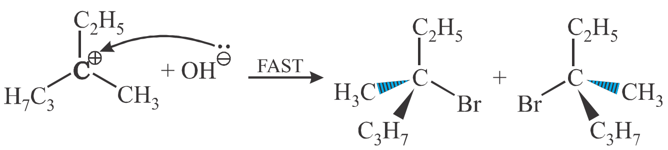
3.2 Substitution Nucleophilic Unimolecular – S_N1

Example - 6

Step 1 :



Step 2 :



Note..

Key Features of S_N1 Mechanism

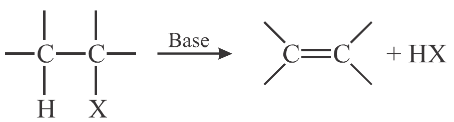
1. Two step reaction. First step is the formation (and rearrangement) of carbocation while second step is the attack of nucleophile on the carbocation.
2. Rate = k [RX]
3. Carbocation is formed.
4. Rearrangement is commonly observed.
5. Racemic mixture is obtained.
6. Order of reactivity of alkyl halides :

$$3° > 2° > 1° > CH_3X$$

   This can be attributed to the stability of the carbocation that is formed.
7. Favoured by protic solvents.

4. ELIMINATION REACTIONS

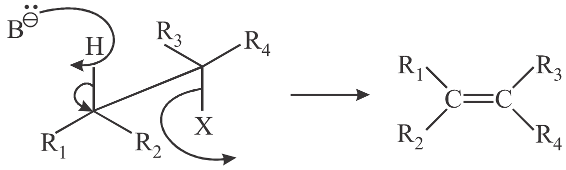
Example - 7



The removal of halogen from the carbon to which it is attached along with the removal of hydrogen from adjacent carbon is called α, β-elimination or simply elimination. Three mechanisms have been outlined for elimination reactions.

4.1 Elimination Bimolecular – E2

Example - 8



Note..

Key Features of E2 Mechanism

1. Single step reaction.
2. Rate = k [RX] [Base]
3. No intermediate is formed. Reaction goes through a transition state.
4. Rearrangement is not observed.
5. Observed in presence of strong bases.
6. Order of reactivity of alkyl halides :

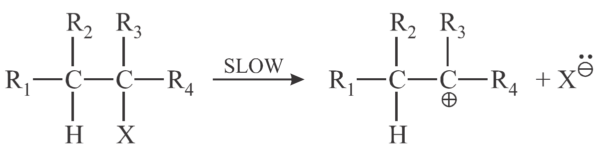
$$3° > 2° > 1°$$

   This can be attributed to the stability of alkene formed.
7. Favoured by aprotic solvents.

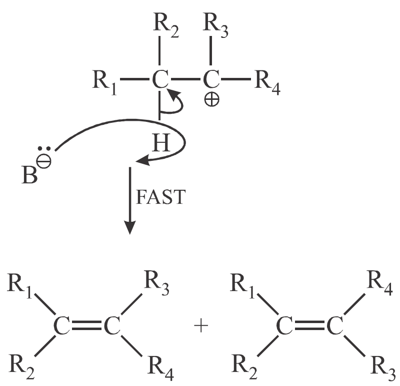
4.2 Elimination Unimolecular – E1

Example - 9

Step 1 :



Step 2 :



Note.. Key Features of E1 Mechanism

1. Two step reaction. First step is the formation of carbocation while second step is the loss of proton by a base.
2. Rate = k [RX]
3. Carbocation is formed.
4. Rearrangement is commonly observed.
5. Observed in presence of weak bases.
6. Order of reactivity of alkyl halides :

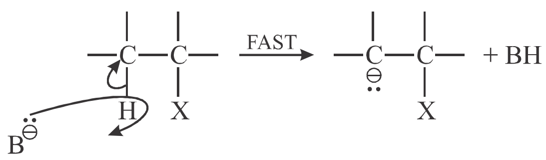
   $$3° > 2° > 1°$$

   This can be attributed to the stability of carbocation as well as the stability of alkene formed.
7. Favoured by protic solvents.

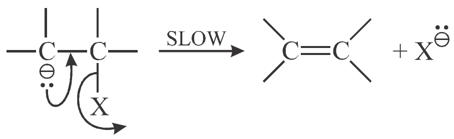
4.3 Elimination Unimolecular via Conjugate Base – E1cB

Example - 10

Step 1 :



Step 2 :

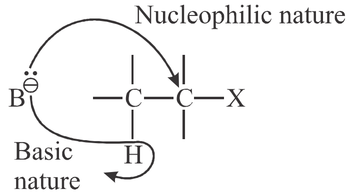


Note.. Key Features of E1cB Mechanism

1. Two step reaction. First step is the formation of carbanion and second step is the loss of leaving group.
2. Rate = k [RX] [Base]
3. Carbanion is formed.
4. Occurs when poor leaving group is present.

5. SUBSTITUTION AND ELIMINATION

Any species that acts as a base can also act as a nucleophile. To understand how elimination and substitution compete with each other, we compare the nucleophilic behaviour with the basic behaviour.



5.1 Nucleophilicity vs Basicity

(a) Nucleophilicity & basicity will be parallel if the comparing nucleophile have same attacking atom e.g.

$$CH_3O^{\ominus} > OH^{\ominus} > CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-O^{\ominus}$$

(b) Negatively charged nucleophiles are stronger than neutral nucleophiles. e.g.

$$OH^{\ominus} > H_2O \quad \text{or} \quad NH_2^{\ominus} > NH_3$$

(c) Electrons on larger atoms are less tightly bound by the nucleus and are more polarisable and more readily available to carbon & will be better nucleophile. But they will be weaker base as their bond with smaller H-atom will be weaker & their conjugate acids will be more reactive. If the attacking atoms are of same size, the stronger bases are better nucleophile. (In a period basicity of anions decreases) e.g.

Acidic Strength :

$$CH_4 < NH_3 < H_2O < HF$$

Basic Strength and Nucleophilicity :

$$^{\ominus}CH_3 > {}^{\ominus}NH_2 > {}^{\ominus}OH > F^{\ominus}$$

(d) If the attacking atoms are different in size, the nucleophilicity depends on the solvents. However, in gaseous phase nucleophilicity parallels basicity.

(e) Nucleophilicity is inversely proportional to stability of anion.

$R-\overset{\overset{\displaystyle O}{\|}}{C}-O^{\ominus}$ is a weaker nucleophile as it is resonance stabilized.

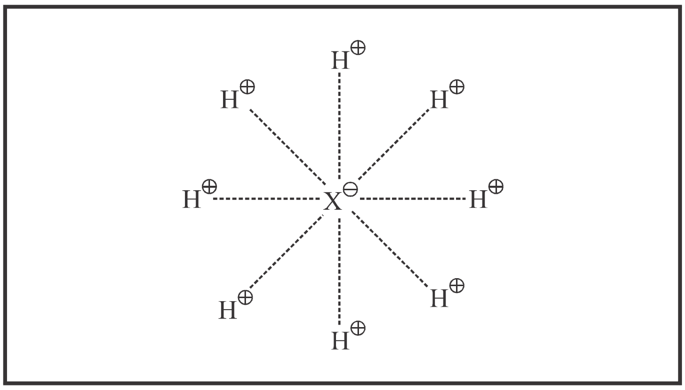
(f) Steric factor limits nucleophilicity

$$CH_3CH_2-O^{\ominus} > CH_3-\underset{\underset{\displaystyle CH_3}{|}}{\overset{\overset{\displaystyle CH_3}{|}}{C}}-O^{\ominus} \quad \text{Nucleophilicity}$$

$$CH_3-\underset{\underset{\displaystyle CH_3}{|}}{\overset{\overset{\displaystyle CH_3}{|}}{C}}-O^{\ominus} > CH_3CH_2-O^{\ominus} \quad \text{Basicity}$$

(g) A strong base can be made a good leaving agent e.g. Oxygen containing group like $OH^{\ominus}$ can be made a weak base in acidic medium by protonation & become a better leaving agent $-\overset{\oplus}{O}H_2$.

(h) **Protic solvent :** These solvents have a hydrogen atom attached to an atom of a strongly electronegative element (e.g. oxygen). Molecules of protic solvent can, therefore form hydrogen bonds to nucleophiles as :



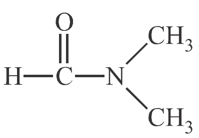
A small nucleophile which is having high charge density than the larger nucleophile is strongly solvated and this solvation hinders the direct approach to the nucleophilic centre. Hence the smaller nucleophile doesn't act as a good nucleophile as the larger one. Hence in protic solvent nucleophilicity is reverse of basicity.

$F^{\ominus}$ $Cl^{\ominus}$ $Br^{\ominus}$ $I^{\ominus}$

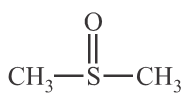
$\longleftarrow$ increasing basicity

$\longrightarrow$ increasing nucleophilicity

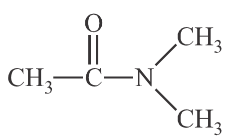
(i) **Aprotic solvents :** These are the polar solvents that don't have H atom, capable of forming H-bonds e.g.



N, N-Dimethyl formamide (DMF)

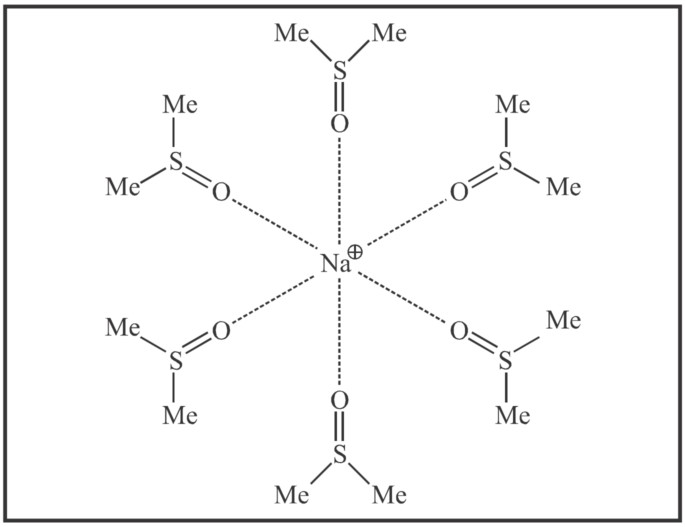


Dimethyl sulphoxide (DMSO)



Dimethyl acetamide (DMA)

These solvents dissolve ionic compounds and solvate the cations.



Now the naked anions are highly reactive as nucleophile and now nucleophilicity follows the basicity e.g.

$F^{\ominus}$ $Cl^{\ominus}$ $Br^{\ominus}$ $I^{\ominus}$

$\longleftarrow$ increasing basicity

$\longleftarrow$ increasing nucleophilicity in aprotic solvents

5.2 Saytzeff vs. Hofmann Rule

When alkene is formed by elimination of alkyl halides, the orientation of the double bond formed is governed by two rules.

5.2.1 Saytzeff's Rule/Zaitsev Rule

This rule suggests the formation of more stable alkene and therefore more substituted double bond. Reactions following this rule are said to be **thermodynamically controlled.**

5.2.2 Hofmann Rule

This rule suggests the formation of less stable alkene and therefore less substituted double bond. In such cases, the more acidic β-hydrogen is abstracted to produce alkene. Such reactions are said to be **kinetically controlled.**

5.3 Effect of Temperature

High temperature favours elimination while low temperature favours substitution reaction.

6. STEREOCHEMISTRY

6.1 Regioselectivity

It is the preference of one direction of chemical bond making or breaking over all other possible directions.

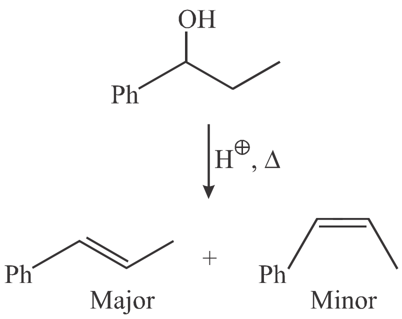
Example - 11

$$CH_3-\underset{\displaystyle |\atop Br}{CH}-CH_2-CH_3 \xrightarrow{\text{alc KOH}} \underset{\text{But-1-ene}}{CH_2=CH-CH_2-CH_3} + \underset{\text{But-2-ene}}{CH_3-CH=CH-CH_3}$$

6.2 Stereoselectivity

Stereoselective reactions give one predominant product because the reaction pathway has a choice. Either the pathway of lower activation energy (kinetic control) is preferred or the more stable product (thermodynamic control).

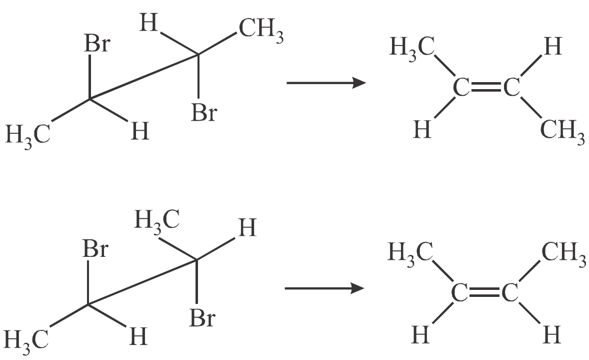
Example - 12



6.3 Stereospecificity

Stereospecific reactions lead to the production of a single isomer as a direct result of the mechanism of the reaction and the stereochemistry of the starting material. There is no choice. The reaction gives a different diastereomer of the product from each stereoisomer of the starting material.

Example - 13

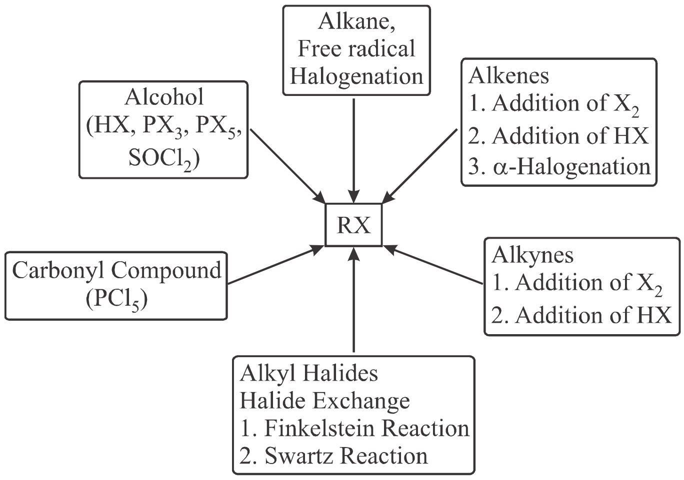


6.4 Chemoselectivity

When there are two or more functional groups in a molecule, a given reagent may react preferentially with one rather than the other. Such reactions are called chemoselective.

7. ALKYL HALIDES

7.1 Preparation of Alkyl Halides



1. Alkanes

$$RH \xrightarrow{Cl_2} RCl + HCl$$

This method gives a mixture of mono, di & trihalides.

2. Alkenes

(i) $R-CH=CH_2 + X_2 \xrightarrow{CCl_4} R-\underset{X}{\underset{|}{CH}}-\underset{X}{\underset{|}{CH_2}}$

(ii) $R-CH=CH_2 + H-X \longrightarrow R-\underset{X}{\underset{|}{CH}}-CH_3$

(iii) (a) $R-CH_2-CH=CH_2 \xrightarrow[400°\text{-}500°C]{X_2} R-\overset{X}{\overset{|}{CH}}-CH=CH_2$

(b) $R-CH_2-CH=CH_2 \xrightarrow{NBS} R-\underset{Br}{\underset{|}{CH}}-CH=CH_2$

3. Alkynes

(i) $R-C\equiv C-H \xrightarrow[CCl_4]{X_2} R-\overset{X}{\underset{X}{C}}-\overset{X}{\underset{X}{C}}-H$

(ii) $R-C\equiv C-H \xrightarrow{2H-X} R-\overset{X}{\underset{X}{C}}-CH_3$

4. Alkyl Halides

(i) Finkelstein Reaction

$$R-Br + NaI \xrightarrow{\text{acetone}} R-I + NaBr$$

$$R-Cl + NaI \xrightarrow{\text{acetone}} R-I + NaCl$$

(ii) Swartz Reaction

$$\begin{matrix} R-I \\ R-Br \\ R-Cl \end{matrix} + AgF \xrightarrow[\text{Polar aprotic}]{\text{DMSO}} R-F + \begin{matrix} AgI \\ AgBr \\ AgCl \end{matrix}$$

5. Alcohol

$$R-OH \xrightarrow{HX} RX + H_2O$$

$$R-OH \xrightarrow{PX_3} RX + H_3PO_3$$

$$R-OH \xrightarrow{PX_5} RX + POX_3 + HX$$

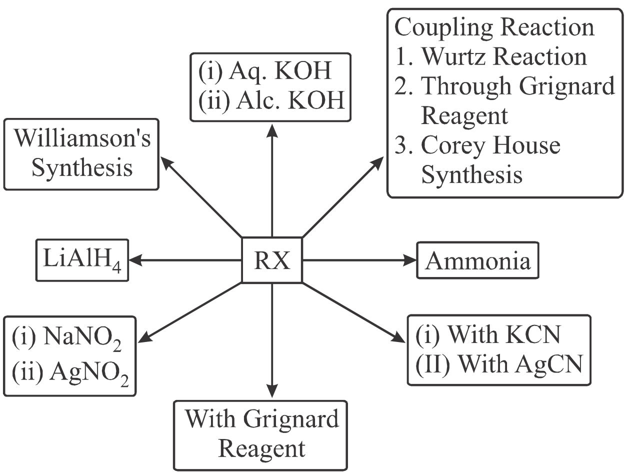
$$R-OH \xrightarrow{SOCl_2/SOBr_2} RX + SO_2 + HX$$

$$R-OH \xrightarrow{SOCl_2,\ \text{Pyridine}} RX + SO_2 + HCl$$

6. Carbonyl Compound

$$R-\overset{O}{\overset{\|}{C}}-H \xrightarrow{PCl_5} R-\overset{Cl}{\underset{Cl}{C}}-H + POCl_3$$

7.2 Reactions of Alkyl Halide



1. Coupling Reaction

(a) Wurtz Reaction

$$2RX + 2Na \xrightarrow{Et_2O} R-R + 2NaX$$

(b) Grignard Reagent

$$R-X + R-MgX \longrightarrow R-R + MgX_2$$

(c) Corey-House Synthesis

$$R-X + 2Li \longrightarrow R-Li + LiX$$

$$2R-Li + CuI \longrightarrow R_2CuLi + LiI$$

$$R_2CuLi + R'-X \longrightarrow R-R' + R-Cu + LiX$$

2. Ammonia

$$R-X + NH_3 \xrightarrow[\Delta]{C_2H_5OH} \underset{1^\circ\ \text{Amine}}{R-NH_2} + HX$$

Note.. If alkyl halide is in excess, then 2° and 3° amines and even quaternary salts are also formed.

$$RX \xrightarrow[\text{alcohol}]{NH_3} \underset{1^\circ\text{Amine}}{RNH_2} \xrightarrow{RX} \underset{2^\circ\text{Amine}}{R_2NH} \xrightarrow{RX} \underset{3^\circ\text{Amine}}{R_3N} \xrightarrow{RX} \underset{\text{(Quaternary salt)}}{R_4\overset{\oplus}{N}X^{\ominus}}$$

This reaction is called Hofmann ammonolysis of alkyl halides.

3. KCN

$$R-X + KCN \longrightarrow R-CN + KX$$

4. AgCN

$$R-X + AgCN \longrightarrow R-\overset{\oplus}{N}\equiv\overset{\ominus}{C}$$

5. $NaNO_2$

$$R-X + NaNO_2 \longrightarrow R-O-N=O + NaX$$

6. $AgNO_2$

$$RX + AgNO_2 \longrightarrow R-NO_2 + AgX$$

7. $LiAlH_4$

$$R-X + LiAlH_4 \longrightarrow R-H$$

8. Williamson's Synthesis

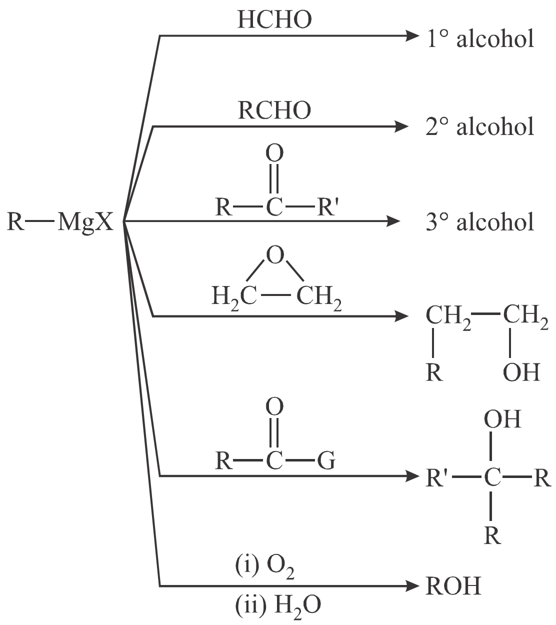
$$R-X + R'-O-Na \longrightarrow R-O-R'$$

9. Aq. KOH & Alc. KOH

$$R-X \xrightarrow{\text{aq. KOH}} R-OH$$

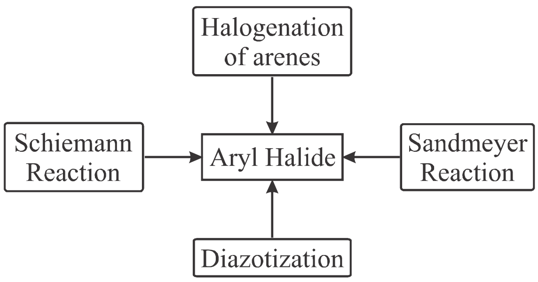
$$R-X \xrightarrow{\text{alc KOH}} \text{Alkene}$$

10. Reactions of R–MgX

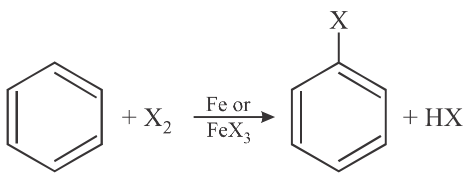


8. ARYL HALIDE/HALOARENES

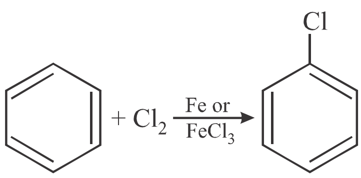
8.1 Preparation of Aryl Halide/Haloarenes



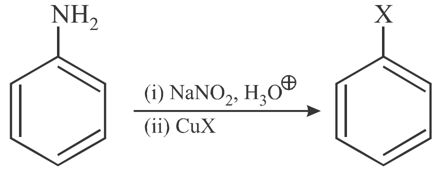
(a) Halogenation of Arenes



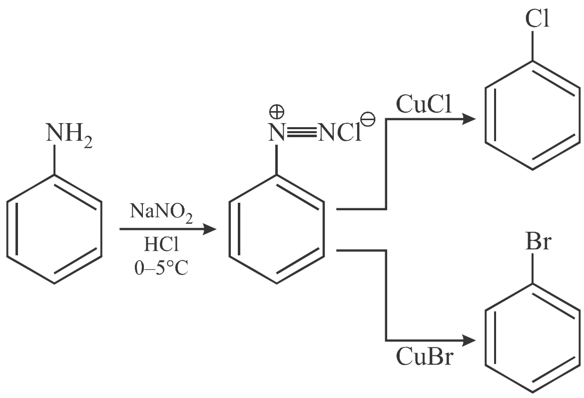
Example - 14



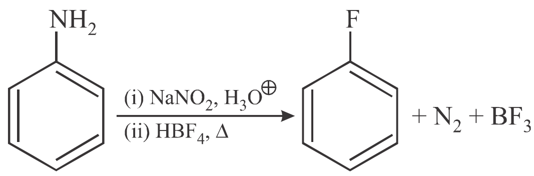
(b) Sandmeyer Reaction



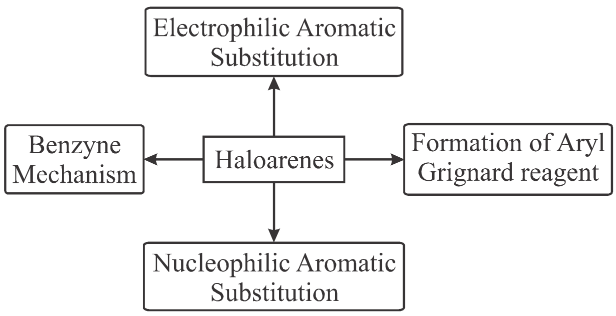
(c) Diazotization



(d) Schiemann Reaction

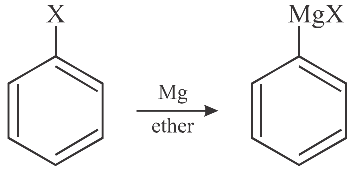


8.2 Reactions of Aryl Halide/Haloarenes



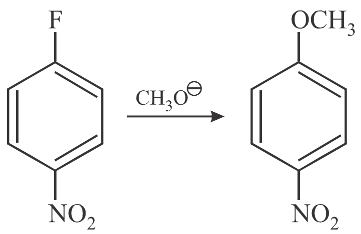
(a) Electrophilic Aromatic Substitution Reaction : Halogens are weakly deactivating and ortho/para directing.

(b) Formation of Aryl Grignard Reagent :



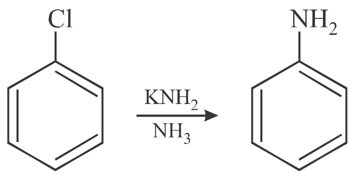
Reactivity order : Ar–I > Ar–Br > Ar–Cl > Ar–F

(c) S_NAr – Aromatic Nucleophilic Substitution Reaction



(d) Benzyne Mechanism (Elimination Addition Mechanism)

Strong bases such as Na, K and amide react readily with aryl halides.



9. REACTIONS OF SPECIAL ALKYL HALIDES

9.1 Di-Halides

$RCHX_2$	Alkylidene Dihalides or Geminal Dihalides
$\begin{array}{cc}RCH & —CH_2\\ \mid & \mid\\ X & X\end{array}$	Alkylene Dihalides or Vicinal (1, 2) Dihalides

9.1.1 Preparation

(a) Alkenes and Alkynes Halogenation of Alkenes & Alkynes

$CH_2{=}CH_2 + X_2 \longrightarrow CH_2X—CH_2X$ Vicinal Dihalide

$CH{\equiv}CH + 2HX \longrightarrow CH_3CHX_2$ Geminal Dihalide

(b) PCl_5 with Diols & Carbonyl Compounds

$$\begin{array}{l}CH_2OH\\ \mid\\ CH_2OH\end{array} + PCl_5 \longrightarrow \begin{array}{l}CH_2Cl\\ \mid\\ CH_2Cl\end{array}$$ Vicinal Dihalide

$CH_3CHO + PCl_5 \longrightarrow CH_3CHCl_2$ Gem Dihalide

9.1.2 Properties

(a) Alcoholic KOH : (Dehydrohalogenation)

$$XCH_2CH_2X \xrightarrow[KOH]{Alcoholic} CH{\equiv}CH$$

$$CH_3CHX_2 \xrightarrow[KOH]{Alcoholic} CH{\equiv}CH$$

(b) Zinc Dust : (Dehalogenation)

$$XCH_2CH_2X \xrightarrow[Alcohol]{Zn} CH_2{=}CH_2$$

$$CH_3CHX_2 \xrightarrow[Alcohol]{Zn} CH_2{=}CH_2$$

(c) Action of aq. KOH : (Alkaline Hydrolysis)

(i) Vicinal Dihalides

$$XCH_2CH_2X \xrightarrow[KOH]{Aqueous} \begin{array}{cc}CH_2 & —CH_2\\ \mid & \mid\\ OH & OH\end{array}$$

Ethylene Glycol

$$CH_3CHXCH_2X \xrightarrow[KOH]{Aqueous} \begin{array}{ccc}CH_3— & CH & —CH_2\\ & \mid & \mid\\ & OH & OH\end{array}$$

Propan-1, 2-diol

(ii) Gem Dihalides

$$CH_3CHX_2 \xrightarrow[KOH]{Aqueous} \underset{\text{(unstable)}}{CH_3CH(OH)_2} \xrightarrow{-H_2O} CH_3CHO$$

$$CH_3CX_2CH_3 \xrightarrow[KOH]{Aqueous} \underset{\text{(unstable)}}{CH_3—C(OH)_2—CH_3}$$

$$\downarrow -H_2O$$

$$CH_3—\overset{\overset{\large O}{\|}}{C}—CH_3$$

Acetone

> **Note..** The above reaction is used to distinguish between gem and vicinal dihalides.

9.2 TRI-HALIDES & TETRA-HALIDES

$CHCl_3$ Chloroform (liquid)	$CHBr_3$ Bromoform (liquid)
$CHCl_3$ Iodoform (yellow solid)	CCl_4 Carbon Tetrachloride (liquid)

9.2.1 CHLOROFORM : $CHCl_3$

(A) Preparation

(i) Ethyl Alcohol : (using NaOH/Cl_2 or $CaOCl_2$)

$$NaOH + Cl_2 \longrightarrow NaOCl + HCl\ ;\ NaOCl \longrightarrow [O]$$

$$C_2H_5OH \xrightarrow[\text{[O]}]{Cl_2} CH_3CHO \xrightarrow{Cl_2} \underset{\text{Chloral}}{CCl_3CHO} + 3HCl$$

Ca (OH)$_2$

$$CCl_3CHO + \underset{\text{Slaked Lime}}{Ca(OH)_2} \longrightarrow \underset{\text{Chloroform}}{2CHCl_3} + Ca(HCOO)_2$$

NaOH

$$CH_3CHO + 3Cl_2 \longrightarrow CCl_3CHO \xrightarrow{\text{Hydrolysis}}$$

$$\underset{\text{Chloralhydrate}}{CCl_3CH(OH)_2} \xrightarrow{NaOH} CHCl_3 + HCOONa + H_2O$$

> **Note..** Pure form of chloroform is prepared from chloral by treating it with NaOH.

(ii) Methyl Ketones

$$\underset{\text{Acetone}}{CH_3COCH_3} + 3Cl_2 \longrightarrow \underset{\text{Trichloroacetone}}{CCl_3COCH_3}$$

$$CCl_3COCH_3 \xrightarrow{Ca(OH)_2} CHCl_3 + (CH_3COO)_2Ca$$

(iii) Carbon Tetrachloride

$$CCl_4 + 2[H] \xrightarrow[\text{HCl}]{Fe + H_2O} CHCl_3 + HCl$$

(iv) Chlorination of Methane at 370ºC

$$CH_4 + 3Cl_2 \xrightarrow[\text{Diffused Sunlight}]{370^\circ C} CHCl_3 + 3HCl$$

(B) Reactions

(i) Oxidation

Chloroform in presence of light and air (O_2) forms a highly posionous gas, Phosgene.

$$2CHCl_3 + O_2 \xrightarrow{\text{light}} \underset{\text{Phosgene}}{2COCl_2} + 2HCl$$

> **Note..** 1% ethanol is added and then chloroform is stored in brown bottles, filled upto brim to stop the above decompostition.

(ii) Carbylamine Reaction

$$RNH_2 + CHCl_3 + 3KOH \rightarrow RNC + 3H_2O + 3\,KCl$$

$$C_6H_5NH_2 + CHCl_3 + 3KOH \rightarrow C_6H_5NC + 3\,H_2O + 3\,KCl$$

Isocyanides (carbylamines) have a very disagreable smell, so the above reaction is used as a test of chlororom and test of 1º (aliphatic and aromatic) amines.

(iii) Hydrolysis

$$CHCl_3 + 3KOH(aq) \longrightarrow HC(OH)_3$$

$$HC(OH)_3 \xrightarrow{-H_2O} HCOOH$$

$$HCOOH \xrightarrow{KOH} HCOOK + H_2O$$

(iv) Formation of Acetylene

$$2CHCl_3 + 6Ag \longrightarrow \underset{\text{Acetylene}}{CH{\equiv}CH} + 6AgCl$$

(v) Formation of Chloropicrin

$$CHCl_3 + HNO_3 \longrightarrow \underset{\substack{\text{Chloropicrin}\\ \text{(an insecticide)}}}{CCl_3NO_2}$$

(vi) Formation of Chloretone

$$CHCl_3 + CH_3COCH_3 \longrightarrow \underset{\substack{\text{Chloretone}\\ \text{(a hypnotic medicine)}}}{(CH_3)_2C(OH){-}CCl_3}$$

9.2.2 IODOFORM : CHI_3

(A) Preparation :

$$C_2H_5OH + 3I_2 + 3Na_2CO_3 \longrightarrow \underset{\text{yellow solid}}{CHI_3}\downarrow + 3HCOONa + 3NaI + 2CO_2\uparrow$$

$$CH_3COCH_3 + 3I_2 + 2Na_2CO_3 \longrightarrow CHI_3\downarrow + 2NaI + CH_3COONa + H_2O + 2CO_2\uparrow$$

> *Note..* This reaction is known as Iodoform reaction or **Iodoform test.** Since the iodoform is a yellow colored solid, so the iodoform reaction is used to test ethyl alcohol, acetaldehyde, sec. alochols of $R(CH_3)CHOH$ (methyl alkyl carbinol) and methyl ketones ($RCOCH_3$), because all these form iodoform. The side product of the iodoform reaction, sodium carboxylate is acidified to produce carboxylic acid (RCOOH).

9.2.3 CARBON TETRACHLORIDE : CCl_4

(A) Preparation :

(i) $\underset{\text{excess}}{CH_4 + 4Cl_2} \xrightarrow[\text{diffused}]{hv} CCl_4 + 4HCl$

(ii) $CS_2 + 3Cl_2 \xrightarrow[Fe/I_2]{AlCl} CCl_4 + S_2Cl_2$

S_2Cl_2 is separated by fractional distillation. It is then treated with more CS_2 to give CCl_4. C washed with NaOH and distilled to obtain pure CCl_4.

$$2S_2Cl_2 + CS_2 \longrightarrow CCl_4 + 6S$$

(iii) $CH_3CH_2CH_3 + Cl_2 \xrightarrow[70\text{-}100\text{ atm}]{400°C} CCl_4 + HCl + C_2Cl_6$

> *Note..* CCl_4 is a colourless and poisonous liquid which is insoluble in H_2O. It is a good solvent for grease and oils. CCl_4 is used in fire extingusher for electric fires as Pyrene. It is also an insecticide for hookworms.

(B) Reactions :

(i) Oxidation

$$CCl_4 + \underset{\text{Steam}}{H_2O} \xrightarrow{500°C} COCl_2 + 2HCl$$

(ii) Reduction

$$CCl_4 + 2[H] \xrightarrow{Fe/H_2O} CHCl_3 + HCl$$

(iii) Hydrolysis

$$CCl_4 + 4KOH(aq) \longrightarrow C(OH)_4 \xrightarrow{2H_2O} CO_2 \xrightarrow{2KOH} K_2CO_3 + H_2O$$

(iv) Action of HF

$$CCl_4 + 4HF \xrightarrow{SbF_6} \underset{\text{Freon}}{CCl_2F_2} + 2HCl$$

9.2.4 VINYL CHLORIDE : $CH_2 = CHCl$

Vinyl group $CH_2 = CH-$

(A) Preparation

(i) $CH{\equiv}CH + HCl \longrightarrow CH_2{=}CHCl$

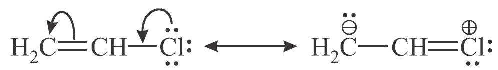
(ii) $\underset{\text{Vicinal Dihalides}}{CH_2ClCH_2Cl} \xrightarrow{KOH(alc.)} CH_2{=}CHCl + KCl + H_2O$

(iii) $CH_2{=}CH_2 + Cl_2 \xrightarrow{600°C} CH_2{=}CHCl + HCl$

(B) Reaction

(i) $CH_2{=}CHCl + \text{alc. KOH} \longrightarrow \underset{\text{Acetylene}}{CH{\equiv}CH} + HCl$

Halogen atom is vinyl chloride is quite stable and does not respond to nucleophilic substitution reactions easily. It is due to resonance stabilisation of vinyl chloride.



9.2.5 ALLYL CHLORIDE : $H_2C = CHCH_2Cl$

(A) Preparation

(i) $\underset{\text{Propene}}{H_2C{=}CHCH_3} + Cl_2 \xrightarrow{500\text{–}600°C} CH_2{=}CHCH_2Cl$

(ii) $\underset{\text{Allyl Alcohol}}{H_2C{=}CHCH_2OH} + PCl_5 \longrightarrow H_2C{=}CHCH_2Cl + POCl_3 + HCl$

(B) Reactions

(i) Addition Reactions

$$CH_2{=}CH{-}CH_2Cl + Cl_2 \longrightarrow CH_2ClCHClCH_2Cl$$

$$CH_2{=}CH{-}CH_2Br + HBr \longrightarrow \underset{\text{1, 2-Dibromopropane}}{CH_3CHBrCH_2Br}$$

The addition follows Markonikov's rule. However in presence of peroxides, 1, 3-dibromopropane is formed.

(ii) Nucleophilic Substitution Reactions

Since in allyl chloride, there is no resonance (unlike in vinyl chloride), nucleophilic substitution reactions take place with ease.

$$CH_2{=}CH{-}CH_2Cl \xrightarrow{KOH(aq)} CH_2{=}CHCH_2OH + KCl$$

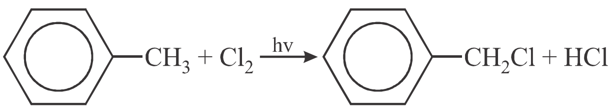
$$CH_2{=}CH{-}CH_2Cl \xrightarrow{NH_3} CH_2{=}CH{-}CH_2NH_2 + HCl$$

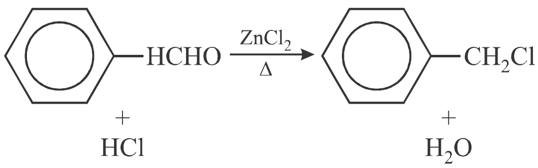
$$CH_2{=}CHCH_2Cl \xrightarrow{KCN} CH_2{=}CH{-}CH_2CN + KCl$$

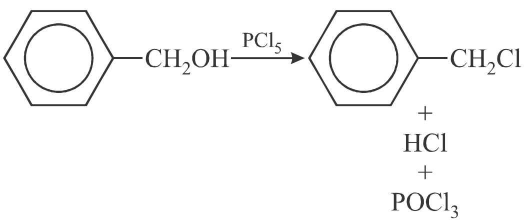
$$CH_2{=}CHCH_2Cl + Mg \xrightarrow[\text{ether}]{\text{dry}} \underset{\text{Grignard Reagent}}{CH_2{=}CHCH_2MgCl}$$

9.2.6 Benzyl Chloride : $C_6H_5CH_2Cl$: $PhCH_2Cl$

(A) Preparation

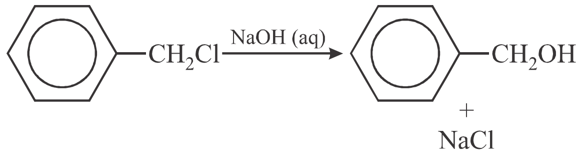
(i) 

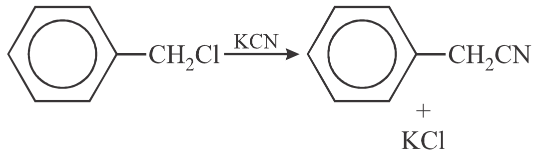
(ii) 

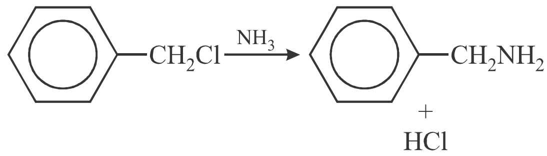
(iii) 

(B) Reactions

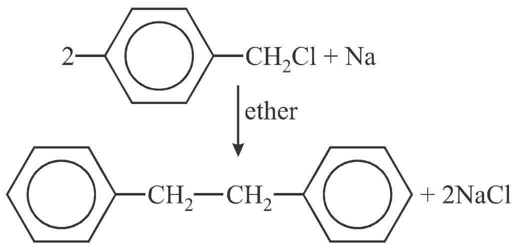
The main reactions are like those of Alkyl Halides (since there is no resonance in benzyl chloride and intermediate benzyl carbonium ion is stable supporting S_N1 substitution). Nucleophillic substitution reactions occur with ease unlike in case of aryl halides (due to resonance in aryl halides).

(i) 

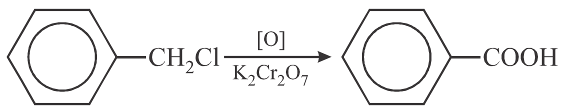
(ii) 

(iii) 

(iv) Wurtz Reaction



(v) Oxidation



10. CHEMISTRY OF GRIGNARD REAGENT : R–Mg–X

10.1 Preparation

$$RX + Mg \xrightarrow[\text{ether}]{\text{reflux in}} \underset{\text{Alkyl Magnesium Halide}}{R{-}Mg{-}X}$$

Note.. In Grignard reagent, we can have phenyl or alkenyl or alkynyl or aromatic group instead of R. All the reactions will remain same.

10.2 Reactions

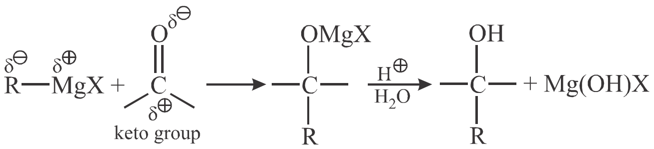
(A) Grignard reagent as a base reacts with compounds containing active H to give alkanes.

$$R{-}MgI + HOH \longrightarrow RH + Mg(OH)\,I$$

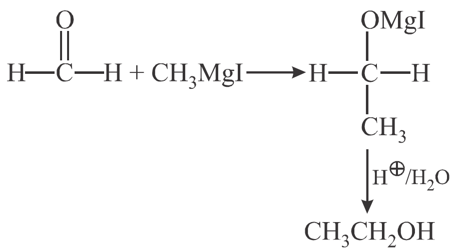
$$R—MgI + R'O—H \longrightarrow RH + Mg(OR')\,I$$

$$R—MgI + R'NH—H \longrightarrow RH + Mg(NHR')\,I$$

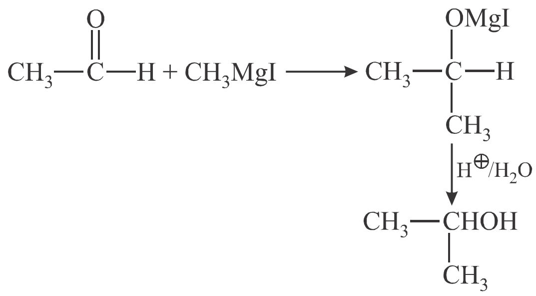
(B) Grignard reagent acts as a strong nucleophile and shows nucleophillic additions to give various products. Alkyl group being electron rich (carbonian) acts as nucleophile in Grignard Reagent.



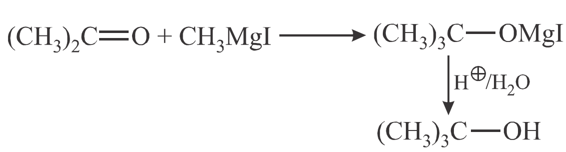
Example - 15



Example - 16



Example - 17



Note...

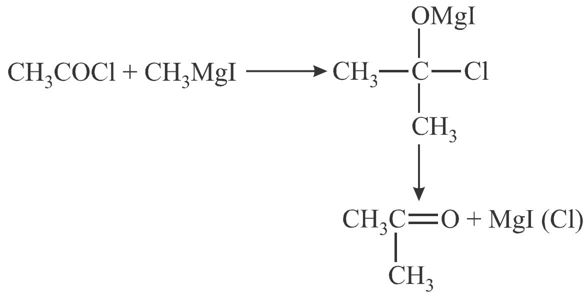
$$HCHO + RMgX \longrightarrow 1^\circ \text{ alcohol}$$

$$R'CHO + RMgX \longrightarrow 2^\circ \text{ alcohol}$$

$$R'COR' + RMgX \longrightarrow 3^\circ \text{ alcohol}$$

(C) Acid Chloride

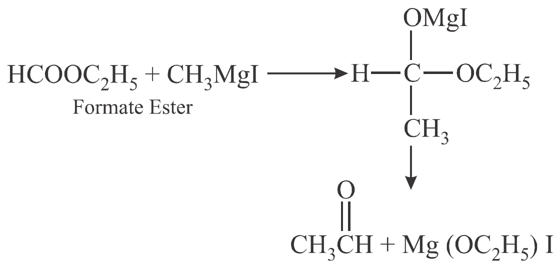
Example - 18



Ketones (acetone) formed further reacts with Grignard reagent to form 3° alcohols (tert. butyl alcohol). However, with 1 : 1 mole ratio of acid halide and Grignard Reagent, one can prepare ketones.

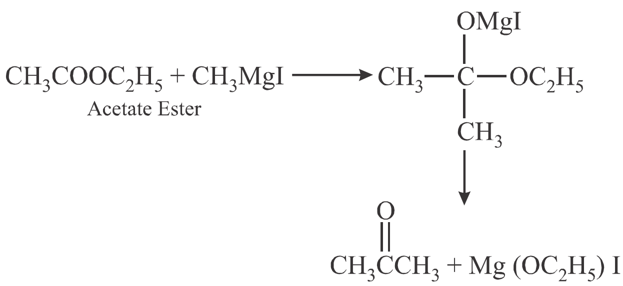
(D) Esters

Example - 19



The aldehydes react further with CH_3MgI to give 2° alcohol, if present in excess. But 1 : 1 mole ratio of reactants will certainly give aldehydes.

Example - 20



The ketones react further with CH_3MgI to give 3° alcohol, if present in excess. But 1 : 1 mole ratio of reactants will certainly give ketones.

(E) Cyanides

Example - 21

$$HC\equiv N + CH_3MgI \longrightarrow CH_3-\overset{\overset{\displaystyle H}{|}}{C}=NMgI \xrightarrow{H^{\oplus}/H_2O} CH_3CHO + \frac{1}{2}N_2 + MgI(OH)$$

Example - 22

$$CH_3C\equiv N + CH_3MgI \longrightarrow (CH_3)_2C=N-MgI \xrightarrow{H^{\oplus}/H_2O} (CH_3)_2C=O + \frac{1}{2}N_2 + MgI(OH)$$

(F) CO_2

$$RMgI + \underset{\text{dry ice}}{O=C=O} \longrightarrow R-\overset{\overset{\displaystyle O}{\|}}{C}-OMgI \xrightarrow{H^{\oplus}/H_2O} \underset{\text{Carboxylic Acid}}{R-\overset{\overset{\displaystyle O}{\|}}{C}-OH} + MgI(OH)$$

(G) Oxygen

$$RMgI + O_2 \xrightarrow[-75^\circ]{\text{ether}} R-O-OMgI \xrightarrow{H^{\oplus}/H_2O} \underset{\text{Alkyl Peroxides}}{R-O-O-H} + MgI(OH)$$

(H) Ethylene Oxide (Oxiranes)

$$CH_3MgI + \underset{\diagdown\ O\ \diagup}{CH_2-CH_2} \longrightarrow CH_3-CH_2-CH_2-OMgI \xrightarrow{H^{\oplus}/H_2O} CH_3CH_2CH_2OH + Mg(OH)I$$

(I) Alkynes

$$CH_3C\equiv C-H + CH_3MgI \longrightarrow CH_3C\equiv C-MgI + CH_4$$

$$CH_3C\equiv C-MgI + CH_3I \longrightarrow CH_3C\equiv C-CH_3 + MgI_2$$

(J) Alkyl Halides

$$R-MgI + CH_3CH_2Br \longrightarrow CH_3CH_2R + Mg(Br)I$$

$$CH_3MgBr + \underset{\text{Alkyl Bromide}}{CH_2=CHCH_2Br} \longrightarrow CH_2CH_2R + MgBr_2$$

(K) Inorganic Halides

$$4C_2H_5MgBr + 2PbCl_2 \longrightarrow \underset{\text{Tetraethyl Lead}}{(C_2H_5)_4Pb} \longrightarrow Pb + 4MgBr(Cl)$$

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Physical and General Properties

1. In CH_3CH_2Br, % of Br is

(a) 80 (b) 73

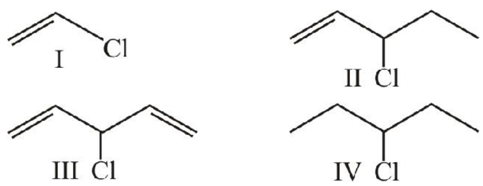
(c) 70 (d) 7

2. Which of the following is a primary halide

(a) Isopropyl iodide (b) Secondary butyl iodide

(c) Tertiary butyl bromide (d) Neo hexyl chloride

3.



Among these compounds, which one has shortest C—Cl bond ?

(a) I (b) II

(c) III (d) IV

4. How many structural isomers of C_3H_6ClBr are possible and how many of them are optically active respectively ?

(a) 5, 2 (b) 5, 3

(c) 4, 2 (d) 3, 2

5. Correct order of B.P. for the alkyl halide is

(a) $C_2H_5Cl > C_2H_5Br > C_2H_5I$

(b) $C_2H_5I > C_2H_5Br > C_2H_5Cl$

(c) $C_2H_5I > C_2H_5Cl > C_2H_5Br$

(d) $C_2H_5Br > C_2H_5I > C_2H_5Cl$

6. The following reaction is known as

$$C_2H_5OH + SOCl_2 \xrightarrow{\text{Pyridine}} C_2H_5Cl + SO_2 + HCl$$

(a) Kharasch effect

(b) Darzen's procedure

(c) Williamson's synthesis

(d) Hunsdiecker synthesis reaction

Preparation of Alkyl Halides

7. Silver acetate + $Br_2 \xrightarrow{CS_2}$. The main product of this reaction is

(a) CH_3 — Br (b) CH_3COI

(c) CH_3COOH (d) None of these

8. Preparation of alkyl halides in laboratory is least preferred by

(a) Treatment of alcohols

(b) Addition of hydrogen halides to alkenes

(c) Halide exchange

(d) Direct halogenation of alkanes

9. Which of the following organic compounds will give a mixture of 1-chlorobutane and 2-chlorobutane on hydrochlorination (HCl)

(a) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$

(b) $HC\equiv C-\overset{\overset{H}{|}}{C}=CH_2$

(c) $CH_2=CH-CH=CH_2$

(d) $CH_2=CH-CH_2-CH_3$

10. Decreasing order of reactivity of HX in the reaction $ROH + HX \rightarrow RX + H_2O$

(a) $HI > HBr > HCl > HF$

(b) $HBr > HCl > HI > HF$

(c) $HCl > HBr > HI > HF$

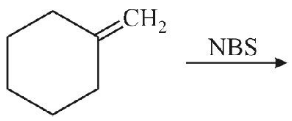
(d) $HF > HBr > HCl > HI$

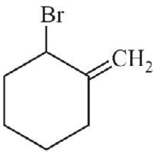
11. $C_3H_8 + Cl_2 \xrightarrow{\text{Light}} C_3H_7Cl + HCl$ is an example of which of the following types of reactions

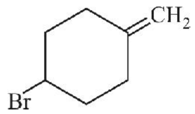
(a) Substitution (b) Elimination

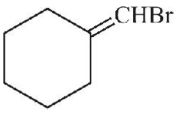
(c) Addition (d) Rearrangement

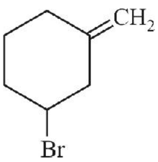
12. What will be the product in the following reaction



(a) 

(b) 

(c) 

(d) 

13. 2, 6 - Dimethylheptane on monochlorination produces……. derivatives

(a) 5 (b) 6

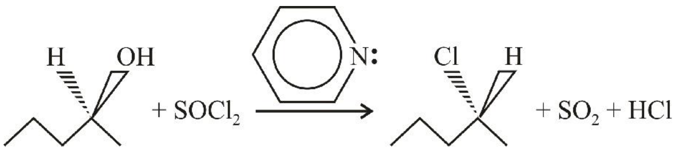
(c) 3 (d) 4

14. Which of the following reagents could be used to convert cyclohexanol to chlorocyclohexane ?

(a) Cl_2, light (b) $SOCl_2$

(c) PBr_3 (d) none of these

15. The reaction



proceed by the mechanism :

(a) S_{N^1} (b) S_{N^2}

(c) S_E (d) S_{N^i}

16. Vinyl chloride reacts with HCl to form

(a) 1, 1- dichloro ethane (b) 1, 2- dichloro ethane

(c) Tetrachloro ethylene

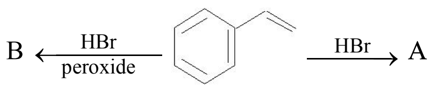
(d) Mixture of 1, 2 and 1, 1 – dichloro ethane

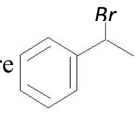
17. $CH_3-CH_2-\underset{Cl}{\underset{|}{CH}}-CH_3$ obtained by chlorination of n-butane will be :

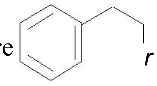
(a) meso form (b) racemic mixture

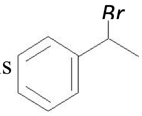
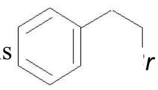
(c) d-form (d) *l*-form

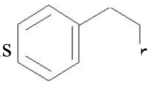
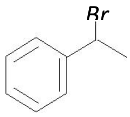
18. Analyse the following reaction and identify the nature of A and B

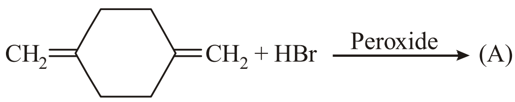


(a) Both A and B are 

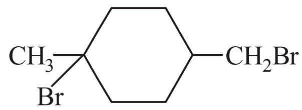
(b) Both A and B are 

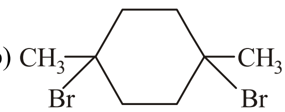
(c) A is  & B is 

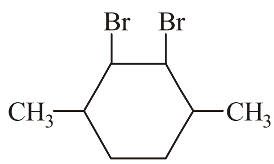
(d) A is  & B is 

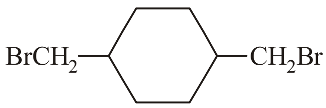
19. 

(A) is :

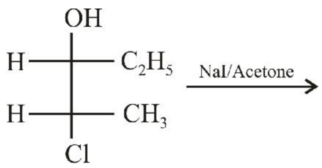
(a) 

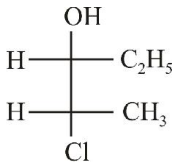
(b) 

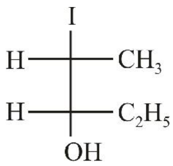
(c) 

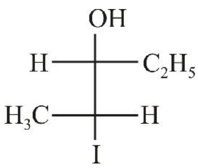
(d) 

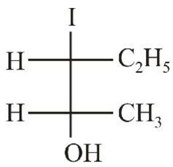
20. In the following reaction find the correct product :



(a) 

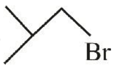
(b) 

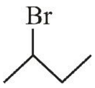
(c) 

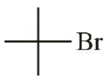
(d) 

21. Formation of free radical takes place with absorption of minimum energy in the formation of :

(a) 

(b) 

(c) 

(d) 

Reactions of Alkyl Halides : Elimination

22. When $CH_3CH_2CHCl_2$ is treated with $NaNH_2$ the product formed is

(a) $CH_3—CH=CH_2$ (b) $CH_3—C\equiv CH$

(c) $CH_3CH_2CH(NH_2)(Cl)$ (d) $CH_3CH_2C(NH_2)_2$

Reactions of Alkyl Halides : Nucleophilic Substitution

23. Ethyl bromide reacts with silver nitrite to form

(a) Nitroethane

(b) Nitroethane and ethyl nitrite

(c) Ethyl nitrite (d) Ethane

24. Which of the following are correct statements about C_2H_5Br

(a) It reacts with metallic Na to give ethane

(b) It gives nitroethane on heating with aqueous ethanolic solution of $AgNO_2$

(c) It gives C_2H_5OH on boiling with alcoholic potash

(d) It forms ethylacetate on heating with silver acetate

25. Ethylene dichloride and ethylidene chloride are isomeric compounds. The false statement about these isomers is that they

(a) React with alcoholic potash and give the same product

(b) Are position isomers

(c) Contain the same percentage of chlorine

(d) Are both hydrolysed to the same product

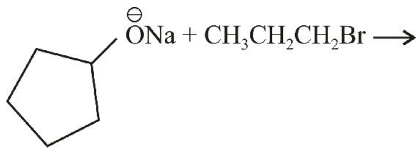
26. Bottles containing C_6H_5I and $C_6H_5CH_2I$ lost their original labels. They were labelled A and B for testing. A and B were separately taken in test tubes and boiled with NaOH solution. The end solution in each tube was made acidic with dilute HNO_3 and then some $AgNO_3$ solution was added. Substance B give a yellow precipitate. Which one of the following statements is true for this experiment

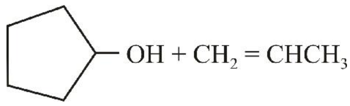
(a) A was C_6H_5I (b) A was $C_6H_5CH_2I$

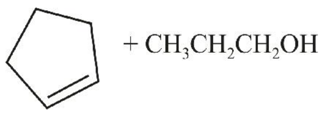
(c) B was C_6H_5I

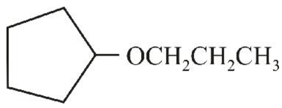
(d) Addition of HNO_3 was unnecessary

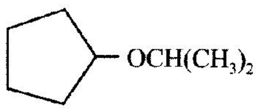
27. What is the major product of the reaction ?



(a) 

(b) 

(c) 

(d) 

28. Which of the following reactions is a good way to prepare methyl tert–butyl ether ?

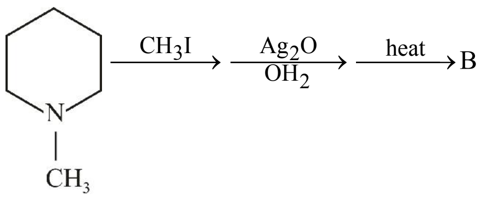
(a) $CH_3O^-Na^+ + (CH_3)_3CBr \longrightarrow$

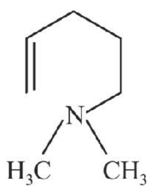
(b) $(CH_3)_3CO^- K^+ + CH_3I \longrightarrow$

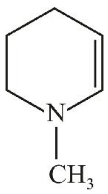
(c) $(CH_3)_3COH + CH_3OH \xrightarrow{NaOH}$

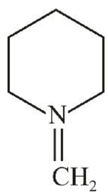
(d) $CH_3MgBr + (CH_3)_3COCl \longrightarrow$

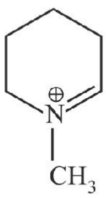
29. The product B of the following sequence, would be

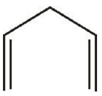


I 

II 

III 

IV 

V 

(a) I
(b) II
(c) III
(d) IV and V

30. Iodine from iodoethane can be substituted by cyanide group in a number of ways. Which of the following is true statement ?

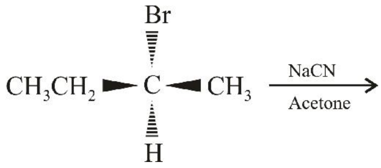
(a) With either KCN or AgCN, EtCN is formed as major product.

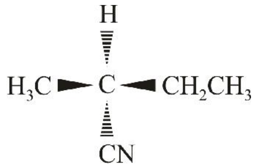
(b) With either KCN or AgCN, EtNC is formed as major product.

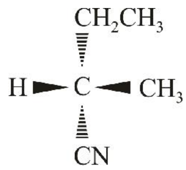
(c) With AgCN, EtCN and with KCN, EtNC is formed.

(d) With AgCN, EtNC while with KCN, EtCN is formed as major product.

31. What is the product of the following reaction ?

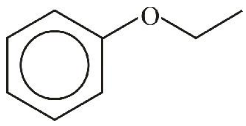


(a) 

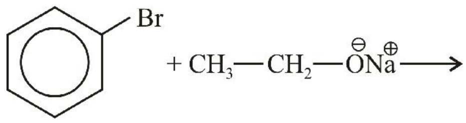
(b) 

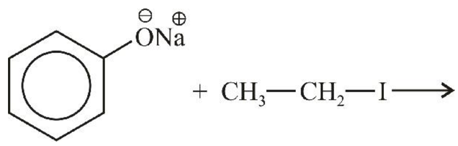
(c) Both (a) and (b)

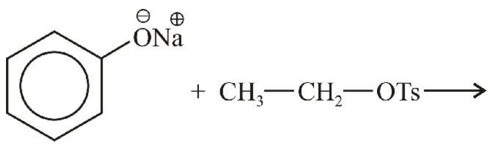
(d) Neither (a) nor (b)

32. 

To make this ether which of the following reactions will be most suitable ?

(a) 

(b) 

(c) 

(d) both (b) and (c)

33. $CH_3-CH_2-CH_2-I \xrightarrow[\text{EtOH}]{\text{Aq.KCN}} \underset{(\text{Major})}{P}$

P should be

(a) $CH_3-CH_2-CH_2-CN$

(b) $CH_3-CH=CH_2$

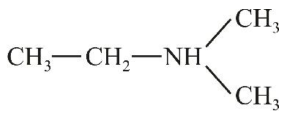
(c) both (a) and (b) in equal proportions

(d) none of these

34. $CH_3-CH_2-NH_2+\underset{(\text{Excess})}{CH_3-I} \longrightarrow \underset{(\text{Major})}{P}$

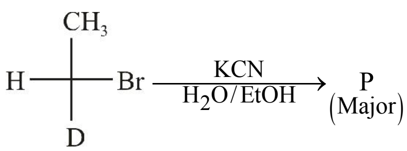
P should be

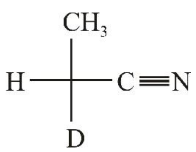
(a) $CH_3-CH_2-NH-CH_3$

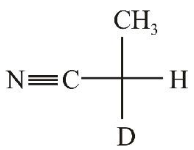
(b) 

(c) $CH_3-CH_2-\overset{\oplus}{N}(CH_3)_3\overset{\ominus}{I}$

(d) $CH_2=CH_2$

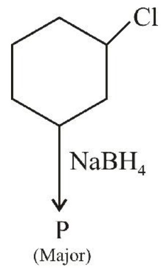
35. 

(a) 

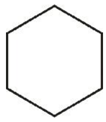
(b) 

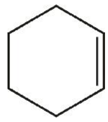
(c) both (a) and (b) in equal proportions

(d) None of these

36. 

P should be

(a) 

(b) 

(c) both of these in equal proportions

(d) none of these

Reactions of Alkyl Halides : Metals

37. An alkyl bromide (X) reacts with Na to form 4, 5-diethyloctane. Compound X is

(a) $CH_3(CH_2)_3Br$

(b) $CH_3(CH_2)_5Br$

(c) $CH_3(CH_2)_3CH.Br.CH_3$

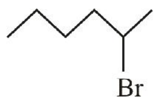
(d) $CH_3(CH_2)_2CH.Br.CH_2CH_3$

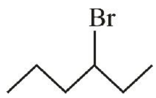
38. $A\,(C_6H_{13}Br) \xrightarrow[2.\,CH_3OH]{1.\,Mg/Ether}$ n–hexane.

$\xrightarrow{\text{Na/Ether}}$ 4, 5–diethyloctane

Deduce the structure of 'A' :

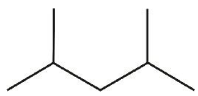
(a) 

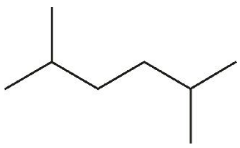
(b) 

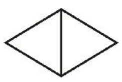
(c) 

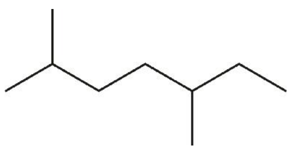
(d) 

39. Which of the following cannot be obtained as the only product in the Wurtz's coupling reaction of halides with sodium metal in ether solution ?

(a) 

(b) 

(c) 

(d) 

40. Corey-House synthesis is a much more convenient method for making alkanes for which of the following reasons ?

(a) Two different alkyl halides can be use. From one alkyl halide, R_2 CuLi is prepared and then this is reacted with another alkyl halide.

(b) The reagent R_2 CuLi can be prepared from 1°, 2° or 3° alkyl halide as well as from halobenzenes

(c) both (a) and (b)

(d) none of these

41. Which of the following combinations of reactants is better for making propane ?

(a) $(CH_3)_2CuLi + CH_3CH_2Br \longrightarrow CH_3CH_2CH_3$

(b) $Et_2CuLi + CH_3-Br \longrightarrow CH_3-CH_2-CH_3$

(c) both of these

(d) none of these

42. Which of the following combinations of reactants is better for making neopentane ?

(a) $(CH_3)_2CuLi + CH_3-C(CH_3)_2-Br \rightarrow CH_3-C(CH_3)_2-CH_3$

(b) $[(CH_3)_3C]_2CuLi + CH_3-Br \rightarrow CH_3-C(CH_3)_2-CH_3$

(c) both (a) and (b)

(d) none of these

43. $CH_3-CH_2-CH_2-Br + CH_3-CH_2-Br \xrightarrow[\text{ether}]{\text{Na}} P$

P is :

(a) P is mixture of three alkanes (n–hexane, n–butane and n–pentane); none is formed in good yield

(b) P is a mixture of only two alkanes (n–hexane and n–butane); both are formed in good yield.

(c) both (a) and (b) depending upon other conditions

(d) none of these

Reactions of Alkyl Halides : Grignard Reagent

44. The addition of the Grignard reagent, CH_3MgBr to acetaldehyde is a nucleophilic addition to the carbonyl group. The nucleophile in this reaction is

(a) CH_3CHO (b) $CH_3^{\oplus}$

(c) $Br^{\ominus}$ (d) $CH_3^{\ominus}$

45. Ester $A(C_4H_8O_2) + \underset{(2\text{ parts})}{CH_3MgBr} \xrightarrow{H_3O^{\oplus}} \underset{(\text{Alcohol B})}{C_4H_{10}O}$

Alcohol B reacts fastest with Lucas reagent. Hence A and B are :

(a) $CH_3-C(=O)-O-C_2H_5$, $(CH_3)_3COH$

(b) $H-C(=O)-O-C_3H_7$, $(CH_3)_2CHOH$

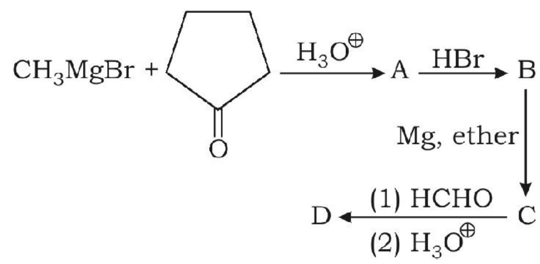
(c) $CH_3-C(=O)-O-C_2H_5$, $(CH_3)_2CHOH$

(d) $H-C(=O)-O-C_3H_7$, $(CH_3)_3COH$

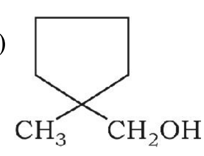
46. $(C_2H_5O)_2CO \xrightarrow[H_3O^{\oplus}]{3CH_3MgBr} A$. A is :

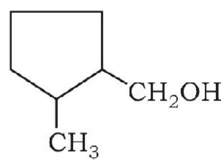
(a) $(C_2H_5O)_2CHOH$ (b) $(CH_3)_3COH$

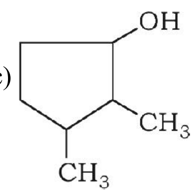
(c) $(C_2H_5)_2C(CH_3)OH$ (d) $(CH_3)_2CHOH$

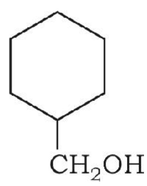
47. 

D is :

(a) 

(b) 

(c) 

(d) 

48. $C_2H_5O-\overset{O}{\overset{\|}{C}}-OC_2H_5 \xrightarrow{2CH_3MgBr}$ A. The product A formed can

(a) give iodoform test

(b) further react with $CH_3MgBr/H_3O^{\oplus}$ to give t-butyl alcohols

(c) be obtained by the ozonolysis of 2, 3 dimethyl–2–butene

(d) all are correct

49. Alkyl halides react with Mg in dry ether to form

(a) Magnesium halide (b) Grignard's reagent

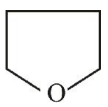
(c) Alkene (d) Alkyne

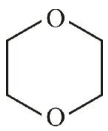
50. When phenyl magnesium bromide reacts with t-butanol, the product would be

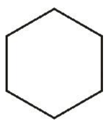
(a) Benzene (b) Phenol

(c) t-butyl benzene (d) t-butyl phenyl ether

51. Which of the following compounds is not suitable solvent for Grignard reagent ?

(a) 

(b) 

(c) 

(d) 

52. $Cl-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow[H_3^+O]{PhMgBr(excess)}$ (X); Product (X) is :

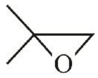
(a) $Ph-\overset{O}{\overset{\|}{C}}-Cl$

(b) $Ph-\overset{O}{\overset{\|}{C}}-Ph$

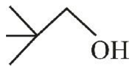
(c) $Ph-\underset{Ph}{\underset{|}{\overset{OH}{\overset{|}{C}}}}-Ph$

(d) $Ph-\overset{O}{\overset{\|}{C}}-O-Ph$

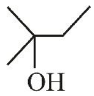
53. In the given reaction

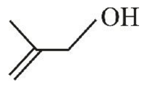
 $\xrightarrow[H^+/H_2O]{H_3CMgBr}$ Product

Find the product of reaction :

(a) 

(b) 

(c) 

(d) 

54. $CH_3-CH-CH_2$ (epoxide, bridged by O) $\xrightarrow[2.\ NH_4Cl]{1.\ CH_3MgBr}$ P

P should be

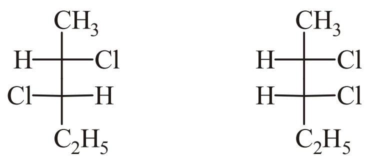
(a) $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2-OH$

(b) $CH_3-\underset{OH}{\underset{|}{CH}}-CH_2-CH_3$

(c) both (a) and (b)

(d) none of these

55. The two optical isomers given below, namely



(a) enantiomers

(b) geometrical isomers

(c) structural isomers

(d) diastereomers

56. Number of stereoisomers of the compound, 2-chloro-4-methylhex-2-ene is

(a) 2 (b) 4

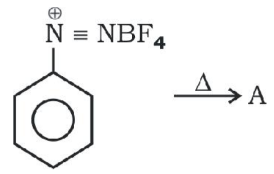
(c) 8 (d) 16

57. The number of enantiomers of the compound $CH_3CHBrCHBrCOOH$ is

(a) 1 (b) 2

(c) 3 (d) 4

Preparation of Aryl Halides

58. 

In the above process product A is

(a) Fluorobenzene (b) Benzene

(c) 1, 4-difluorobenzene (d) 1, 3-difluorobenzene

59. C_6H_5Cl prepared by aniline with

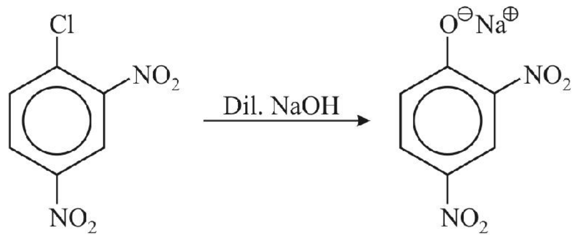
(a) HCl (b) Cu_2Cl_2

(c) Cl_2 in presence of anhydrous $AlCl_3$

(d) HNO_2 and then heated with Cu_2Cl_2

Reactions of Aryl Halides

60. The following transformation proceeds through

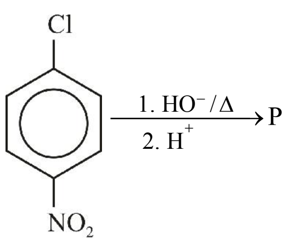


(a) benzyne intermediate

(b) oxirane

(c) electrophilic - addition

(d) nucleophilic aromatic substitution

61. 

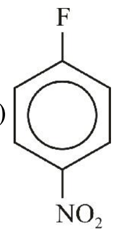
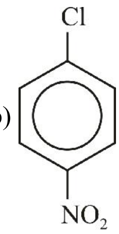
This reaction follows

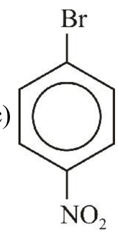
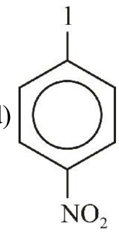
(a) S_{N^1} pathway (b) S_{N^2} pathway

(c) addition-elimination pathway

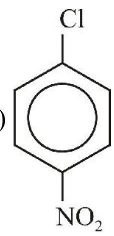
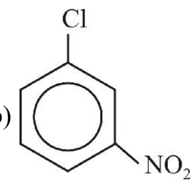
(d) none of these

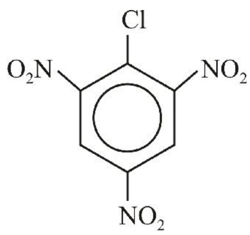
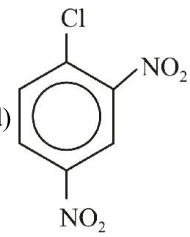
62. In addition-elimination pathway of nucleophilic aromatic substitutions, which of the following compounds is most reactive ?

(a)  (b) 

(c)  (d) 

63. In addition-elimination pathway of nucleophilic aromatic substitutions, which of the following compounds is most reactive ?

(a)  (b) 

(c)  (d) 

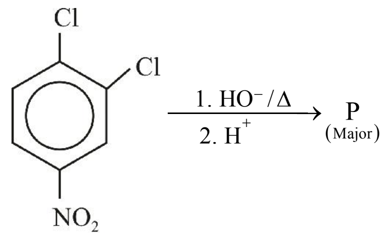
64. In addition elimination pathway of nucleophilic aromatic substitutions, a–NO_2 groups is

(a) o, p–directing and activating

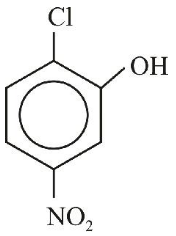
(b) o, p–directing and deactivating

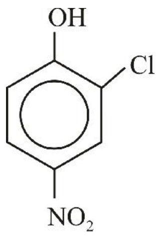
(c) m–directing and activating

(d) m–directing and deactivating

65. 

Based on the above reacation, the major product would be

(a) 

(b) 

(c) both in equal proportions (d) none of these

66. Which of the following has the fastest rate of the reaction with $NaOC_2H_5$?

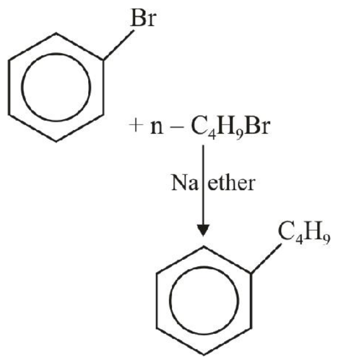
(a) p–fluoronitrobenzene (b) p–bromonitrobenzene
(c) p–chloronitrobenzene (d) p–iodonitrobenzene

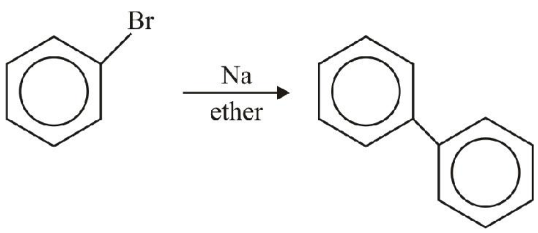
67. How many isomers are possible with trisubstituted benzene with all the three groups same ?

(a) 3 (b) 4
(c) 5 (d) 6

68. Which of the following reaction is called Wurtz-Fittig reaction ?

(a) $n\text{-}C_4H_9Br \xrightarrow[\text{ether}]{Na} C_4H_9 - C_4H_9$

(b) 

(c) 

(d) both (b) and (c)

Polyhalogen Compounds

69. Iodoform can be used as

(a) Anaesthetic (b) Antiseptic
(c) Analgesic (d) Antifebrin

70. Which of the following is an anaesthetic

(a) C_2H_4 (b) $CHCl_3$
(c) CH_3Cl (d) C_2H_5OH

71. An important insecticide is obtained by the action of chloral on chlorobenzene. It is

(a) BHC (b) Gammexene
(c) DDT (d) Lindane

72. Statement "Ozone in atmosphere is decreased by chloro-fluoro-carbon (Cl_2F_2C)

(a) Is true
(b) Is false
(c) Only in presence of CO_2
(d) Only in absence of CO_2

73. DDT is prepared by reacting chlorobenzene with

(a) CCl_4 (b) $CCl_3—CHO$
(c) $CHCl_3$ (d) Ethane

74. A sample of chloroform being used as anaesthetic is tested by

(a) Fehling solution
(b) Ammonical Cu_2Cl_2
(c) $AgNO_3$ solution
(d) $AgNO_3$ solution after boiling with alcoholic KOH solution

75. Chloroform for anesthetic purposes is tested for its purity with the reagent

(a) Silver nitrate
(b) Lead nitrate
(c) Ammoniacal Cu_2Cl_2 (d) Lead nitrate

76. Chloropicrin is obtained by the reaction of

(a) Chlorine on picric acid (b) Nitric acid on chloroform
(c) Steam on carbon tetrachloride
(d) Nitric acid on chlorobenzene

77. Haloforms are trihalogen derivatives of

(a) Ethane (b) Methane
(c) Propane (d) Benzene

78. Number of π–bonds present in B.H.C. (Benzene hexachloride) are

(a) 6 (b) Zero
(c) 3 (d) 12

79. CCl_4 cannot give precipitate with $AgNO_3$ due to

(a) Formation of complex with $AgNO_3$

(b) Evolution of Cl_2 gas

(c) Chloride ion is not formed

(d) $AgNO_3$ does not give silver ion

80. The formula of freon-12 is :

(a) $CClF_3$ (b) CH_2Cl_2

(c) CCl_2F_2 (d) CH_2F_2

Conversions/Roadmaps

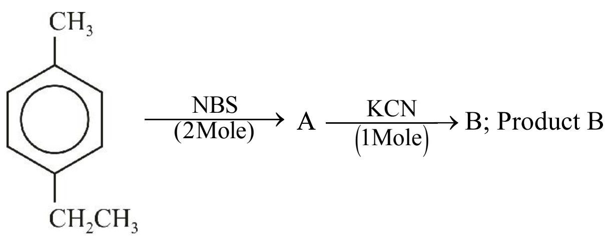
81. In the following sequence of reactions

$$CH_3CH_2CH_2Br \xrightarrow{KOH(alc)} (A) \xrightarrow{HBr} (B) \xrightarrow{KOH(aq.)} (C),$$

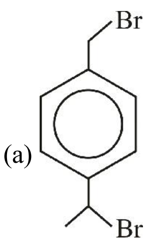
The product (C) is

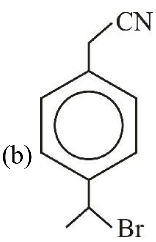
(a) Propan – 2 – *ol* (b) Propan – 1 – *ol*

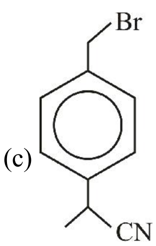
(c) Propyne (d) Propene

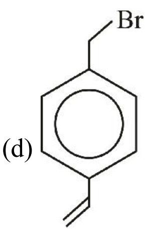
82. 

is :

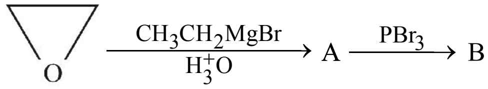
(a) 

(b) 

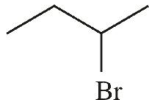
(c) 

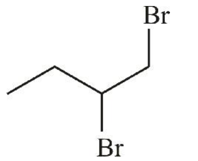
(d) 

83. Consider the following sequence of reaction

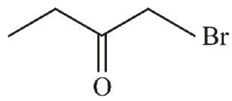


The product B is :

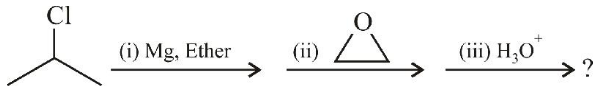
(a) 

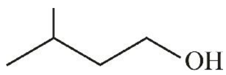
(b) 

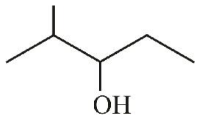
(c) 

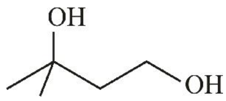
(d) 

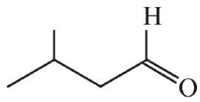
84. What is the major product of the reaction sequence ?



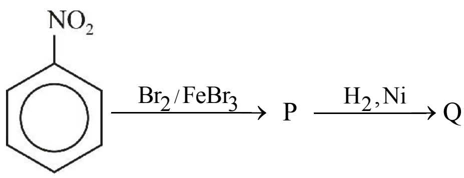
(a) 

(b) 

(c) 

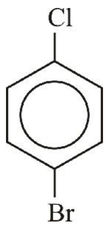
(d) 

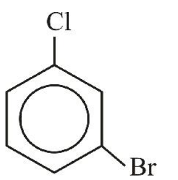
85. Consider the following reactions,

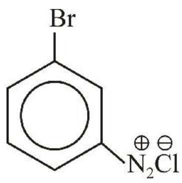


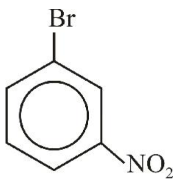
$$\xrightarrow[0°–5°C]{NaNO_2+HCl} R \xrightarrow{CuCl} S$$

The end product 'S' is :

(a) 

(b) 

(c) 

(d) 

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

1. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be **(2003)**

(a) propane (b) pentane

(c) isopentane (d) neopentane

2. The compound formed on heating chlorobenzene with chloral in the presence of concentrated sulphuric acid is **(2004)**

(a) gammexane (b) DDT

(c) freon (d) hexachloroethane

3. Reaction of one molecule of HBr with one molecule of 1, 3-butadiene at 40°C gives predominantly **(2005)**

(a) 1-bromo-2-butene under kinetically controlled conditions

(b) 3-bromobutene under thermodynamically controlled conditions

(c) 1-bromo-2-butene under thermodynamically controlled conditions

(d) 3-bromobutene under kinetically controlled conditions

4. Elimination of bromine from 2-bromobutane results in the formation of **(2005)**

(a) predominantly 2-butyne (b) predominantly 1-butene

(c) predominantly 2-butene

(d) equimolar mixture of 1 and 2-butene

5. Of the four isomeric hexanes, the isomer which can give two monochlorinated compounds, is **(2005)**

(a) 2-methylpentane (b) 2, 2-dimethylbutane

(c) 2, 3-dimethylbutane (d) n-hexane

6. 2-methyl butane on reacting with bromine in the presence of sunlight gives mainly **(2005)**

(a) 1-bromo-3-methylbutane (b) 2-bromo-3-methylbutane

(c) 2-bromo-2-methylbutane (d) 1-bromo-2-methylbutane

7. Alkyl halides react with dialkyl copper reagents to give **(2005)**

(a) alkenyl halides (b) alkanes

(c) alkyl copper halides (d) alkenes

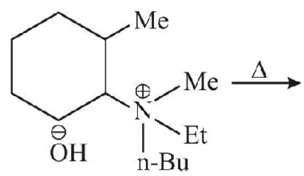
8. $CH_3Br + Nu^- \rightarrow CH_3-Nu + Br^-$ **(2006)**

The decreasing order of the rate of the above reaction with nucleophile (Nu^-) A and D is

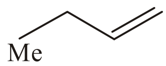
$[Nu = (A)\ PhO^-, (B)\ AcO^-, (C)\ HO^-, (D)\ CH_3O^-]$

(a) $D > C > A > B$ (b) $D > C > B > A$

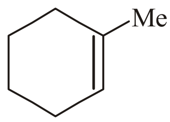
(c) $A > B > C > D$ (d) $B > D > C > A$

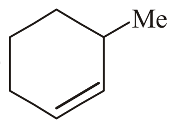
9. 

The alkene formed as a major product in the above elimination reaction is **(2007)**

(a) 

(b) $CH_2=CH_2$

(c) 

(d) 

10. Trans-2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces **(2006)**

(a) 4-phenylcyclopentene (b) 2-phenylcyclopentene

(c) 1-phenylcyclopentene (d) 3-phenylcyclopentene

11. HBr reacts with $CH_2=CH-\ddot{\underset{\cdot\cdot}{O}}CH_3$ under anhydrous conditions at room temperature to give **(2006)**

(a) CH_3CHO and CH_3Br

(b) $BrCH_2CHO$ and CH_3OH

(c) $BrCH_2-CH_2-OCH_3$

(d) $H_3C-CHBr-OCH_3$

12. Which of the following reactions will yield 2, 2-dibromopropane ? **(2007)**

(a) $CH_3-C\equiv CH + 2HBr \longrightarrow$

(b) $CH_3CH=CHBr + HBr \longrightarrow$

(c) $CH\equiv CH + 2HBr \longrightarrow$

(d) $CH_3-CH=CH_2 + HBr \longrightarrow$

13. The treatment of CH_3MgX with $CH_3C\equiv C-H$ produces **(2008)**

(a) $CH_3-CH=CH_2$ (b) $CH_3C\equiv C-CH_3$

(c) $CH_3-\overset{H}{\overset{|}{C}}=\overset{H}{\overset{|}{C}}-CH_3$ (d) CH_4

14. The organic chloro compound, which shows complete stereochemical inversion during an S_N2 reaction is **(2008)**

(a) $(C_2H_5)_2CHCl$ (b) $(CH_3)_3CCl$

(c) $(CH_3)_2CHCl$ (d) CH_3Cl

15. Following reaction **(2002)**

$(CH_3)_3CBr + H_2O \rightarrow (CH_3)_3COH + HBr$ is an example of

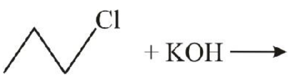
(a) elimination reaction (b) free radical substitution

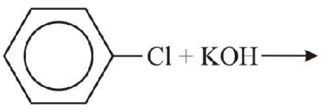
(c) nucleophilic substitution

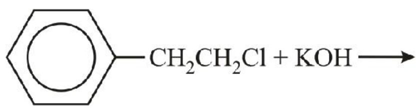
(d) electrophilic substitution

16. S_N1 reaction is feasible in **(2002)**

(a) 

(b) 

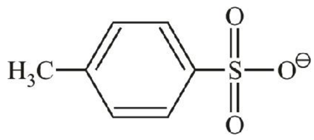
(c) 

(d) 

17. The decreasing order of nucleophilicity among the nucleophiles **(2005)**

(A) $CH_3\underset{\underset{O}{\|}}{C}-O^{\ominus}$ (B) $CH_3O^{\ominus}$

(C) $CN^{\ominus}$

(D) 

(a) (C), (B), (A), (D) (b) (B), (C), (A), (D)

(c) (D), (C), (B), (A) (d) (A), (B), (C), (D)

18. Tertiary alkyl halides are practically inert to substitution by S_N2 mechanism because of **(2005)**

(a) steric hindrance (b) inductive effect

(c) instability (d) insolubility

19. Which of the following is the correct order of decreasing S_N2 reactivity ? **(2007)**

(a) $RCH_2X > R_3CX > R_2CHX$

(b) $RCH_2X > R_2CHX > R_3CX$

(c) $R_3CX > R_2CHX > RCH_2X$

(d) $R_2CHX > R_3CX > RCH_2X$

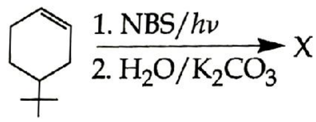
(X = a halogen)

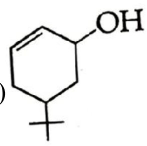
20. The major organic compound formed by the reaction of 1, 1, 1– trichloroethane with silver powder is : **(2014)**

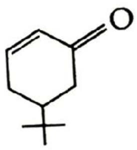
(a) Ethene (b) 2-Butyne

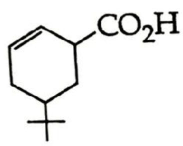
(c) 2-Butene (d) Acetylene

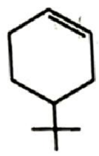
21. The product of the reaction given below is : **(2016)**



(a) 

(b) 

(c) 

(d) 

22. The reaction of propene with HOCl ($Cl_2 + H_2O$) proceeds through the intermediate : **(2016)**

(a) $CH_3 - CH^+ - CH_2 - Cl$

(b) $CH_3 - CH(OH) - CH_2^+$

(c) $CH_3 - CHCl - CH_2^+$

(d) $CH_3 - CH^+ - CH_2 - OH$

23. The increasing order of reactivity of the following halides for the S_N1 reaction is **(2017)**

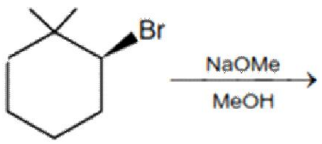
(I) $CH_3CH(Cl)CH_2CH_3$ (II) $CH_3CH_2CH_2Cl$

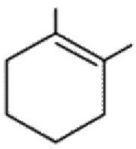
(III) p-$H_3CO–C_6H_4 – CH_2Cl$

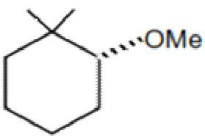
(a) (III) < (II) < (I) (b) (II) < (I) < (III)

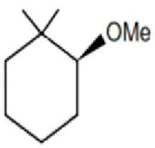
(c) (I) < (III) < (II) (d) (II) < (III) < (I)

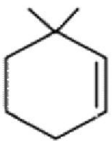
24. The major product of the following reaction is : **(2018)**



(a) 

(b) 

(c) 

(d) 

JEE MAINS ONLINE QUESTION

1. For the compounds

CH_3Cl, CH_3Br, CH_3I and CH_3F, the correct order of increasing C – halogen bond length is:

Online 2014 SET (1)

(a) $CH_3F < CH_3Cl < CH_3Br < CH_3I$

(b) $CH_3F < CH_3Br < CH_3Cl < CH_3I$

(c) $CH_3F < CH_3I < CH_3Br < CH_3Cl$

(d) $CH_3Cl < CH_3Br < CH_3F < CH_3I$

2. In a nucleophilic substitution reaction:

$R - Br + Cl^- \xrightarrow{DMF} R - Cl + Br^-$,

Which one of the following undergoes complete inversion of configuration ? **Online 2014 SET (1)**

(a) $C_6H_5CHC_6H_5Br$ (b) $C_6H_5CH_2Br$

(c) $C_6H_5CH_2CH_3Br$ (d) $C_6H_5CCH_3C_6H_5Br$

3. Conversion of benzene diazonium chloride to chloro benzene is an example of which of the following reactions?

Online 2014 SET (3)

(a) Claisen (b) Friedel-craft

(c) Sandmeyer (d) Wurtz

4. In the presence of peroxide, HC*l* and HI do not give anti-Markownikoff's addition to alkenes because:

Online 2014 SET (3)

(a) HCl is oxidizing and the HI is reducing

(b) All the steps are exothermic in HC*l* and HI

(c) Both HCl and HI are strong acids

(d) One of the steps is endothermic in HCl and HI

5. The major product formed when 1, 1, 1 trichloro–propane is treated with aqueous potassium hydroxide is :

Online 2014 SET (4)

(a) 2–Propanol (b) Propyne

(c) Propionic acid (d) 1–propanol

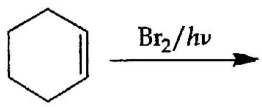
6. The gas evolved on heating CH_3MgBr in methanol is :

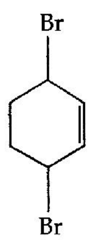
Online 2016 SET (1)

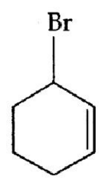
(a) HBr (b) Methane

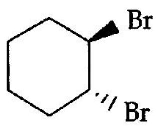
(c) Ethane (d) Propane

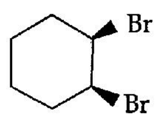
7. Bromination of cyclohexene under conditions given below yields : **Online 2016 SET (2)**



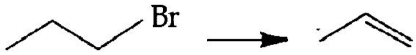
(a) 

(b) 

(c) 

(d) 

8. Which one of the following reagents is **not** suitable for the elimination reaction ? **Online 2016 SET (2)**



(a) $NaOH / H_2O$ (b) NaOEt / EtOH

(c) $NaOH / H_2O - EtOH$

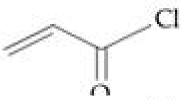
(d) NaI

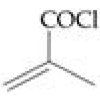
9. Fluorination of an aromatic ring is easily accomplished by treating a diazonium salt with HBF_4. Which of the following conditions is correct about this reaction ? **Online 2016 SET (2)**

(a) Only heat (b) $NaNO_2 / Cu$

(c) Cu_2O / H_2O (d) NaF/Cu

10. Which of the following compounds will not undergo Friedel Craft's reaction with benzene ? **Online 2017 SET (1)**

(a)  (b) 

(c)  (d) 

11. The major product of the following reaction is :

$$\underset{\displaystyle Br}{CH_3CH}\underset{\displaystyle Br}{CH_2CH}CH_2CH_3 \xrightarrow[\text{heat}]{KOH,\ CH_3OH}$$

Online 2017 SET (1)

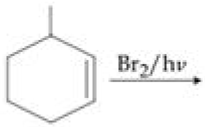
(a) $CH_2=CHCH_2CH=CHCH_3$

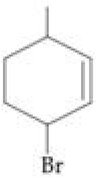
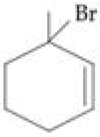
(b) $CH_2=CHCH=CHCH_2CH_3$

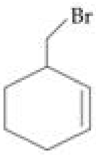
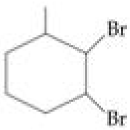
(c) $CH_3CH=C=CHCH_2CH_3$

(d) $CH_3CH=CH-CH=CHCH_3$

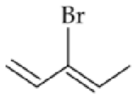
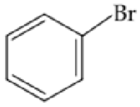
12. The major product of the following reaction is **Online 2017 SET (2)**

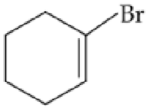
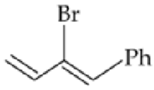


(a)  (b) 

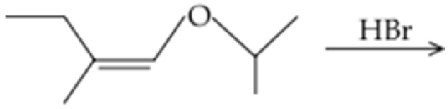
(c)  (d) 

13. Which of the following will most readily given the dehydrohalogenation product. **Online 2018 SET (1)**

(a)  (b) 

(c)  (d) 

14. The total number of optically active compounds formed in the following reaction is : **Online 2018 SET (2)**

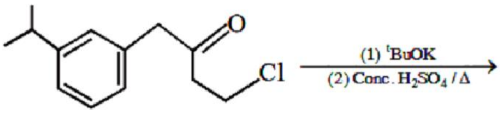


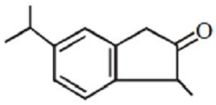
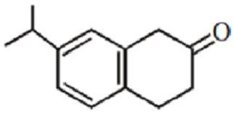
(1) Two (2) Four

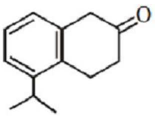
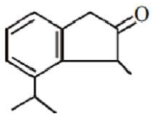
(3) Six (4) Zero

15. The major product of the following reaction is :

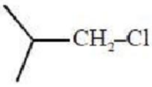
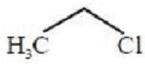
Online 2019 SET (2)

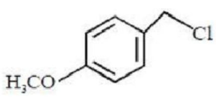
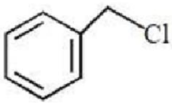


(a)  (b) 

(c)  (d) 

16. Increasing order of reactivity of the following compounds for S_N1 substitution is : **Online 2019 SET (3)**

(A)  (B) 

(C)  (D) 

(a) (B) < (C) < (D) < (A) (b) (A) < (B) < (D) < (C)

(c) (B) < (A) < (D) < (C) (d) (B) < (C) < (A) < (D)

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked **"S"** are single choice questions
2. All questions marked **"M"** are multiple choice questions
3. All questions marked **"C"** are comprehension based questions
4. All questions marked **"A"** are assertion–reason type questions

   (A) If both assertion and reason are correct and reason is the correct explanation of assertion.

   (B) If both assertion and reason are true but reason is not the correct explanation of assertion.

   (C) If assertion is true but reason is false.

   (D) If reason is true but assertion is false.
5. All questions marked **"X"** are matrix–match type questions
6. All questions marked **"I"** are integer type questions

Physical and General Properties

1. (S) Which of the following is liquid at room temperature ?

(a) CH_3I (b) CH_3Br

(c) C_2H_5Cl (d) CH_3F

2. (S) Which of the following represents the correct order of densities ?

(a) $CCl_4 > CHCl_3 > CH_2Cl_2 > CH_3Cl > H_2O$

(b) $CCl_4 > CHCl_3 > CH_2Cl_2 > H_2O > CH_3Cl$

(c) $H_2O > CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4$

(d) $CCl_4 > CHCl_3 > H_2O > CH_2Cl_2 > CH_3Cl$

3. (S) Which of the following has the highest normal boiling point ?

(a) iodobenzene (b) bromobenzene

(c) chlorobenzene (d) fluorobenzene

4. (S) Which of the following are arranged in the decreasing order of dipole moment ?

(a) CH_3Cl, CH_3Br, CH_3F

(b) CH_3Cl, CH_3F, CH_3Br

(c) CH_3Br, CH_3Cl, CH_3F

(d) CH_3Br, CH_3F, CH_3Cl

5. (A) **Assertion :** The carbon halogen bond in an aryl halide is shorter than the carbon halogen bond in an alkyl halide.

Reason : A bond formed of an sp^3 orbital should be shorter than the corresponding bond involving an sp^2 orbital.

(a) A (b) B

(c) C (d) D

Preparation of Alkyl Halides

6. (S) The conversion of 2, 3-dibromobutane to 2-butene with Zn and alcohol is

(a) β-Elimination

(b) Redox reaction

(c) Both β-elimination and redox reaction

(d) α-Elimination

7. (A) **Assertion :** Alkyl iodide can be prepared by treating alkyl chloride/bromide with NaI in acetone.

Reason : NaCl/NaBr are soluble in acetone while NaI is not.

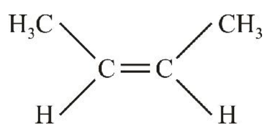
(a) A (b) B

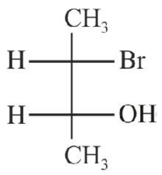
(c) C (d) D

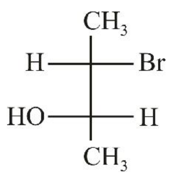
8. (S) The number of possible enantiomeric pairs that can be produced during monochlorination of 2-methyl butane is :

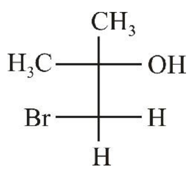
(a) 3 (b) 4

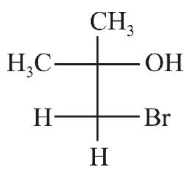
(c) 2 (d) 1

9. (S)  $\xrightarrow{Br_2+H_2O}$ Product :

(a) 

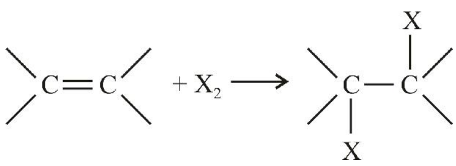
(b) 

(c) 

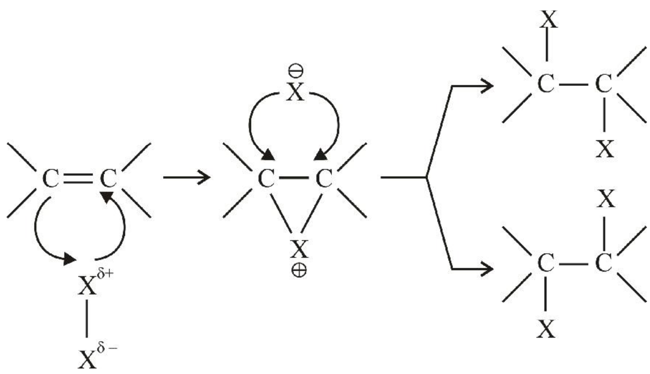
(d) 

Comprehension

Addition of X_2 on alkene is electrophilic addition reaction. Reaction proceed through the formation of 3-membered cyclic halonium ion. Nucleophile X^- attacks from backside of cyclic halonium ion hence total reaction is anti addition reaction. If this reaction proceed in polar solvent then solvent itself acts as nucleophile.



Mechanism :



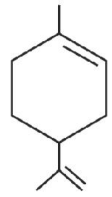
10. (C) Which of the following statements is incorrect ?

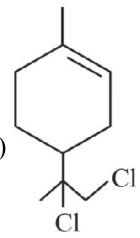
(a) Symmertical trans alkene gives 2 products on reaction with Br_2/CCl_4

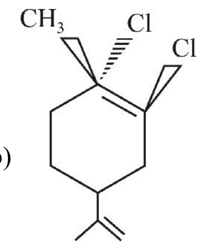
(b) Symmetrical cis alkene gives 2 products on reaction wit Br_2/CCl_4

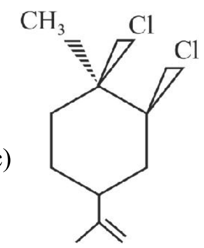
(c) Trans alkenes give erythro product

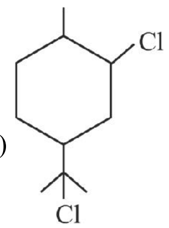
(d) Cis alkenes gives threo product

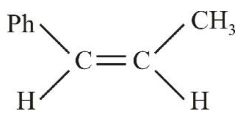
11. (C)  $\xrightarrow[CCl_4]{Cl_2(1Mole)}$

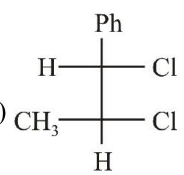
(a) 

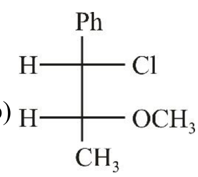
(b) 

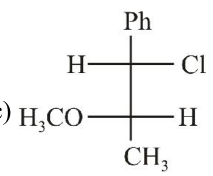
(c) 

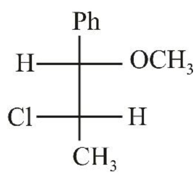
(d) 

12. (C)  $\xrightarrow{Cl_2, CH_3OH}$ Major product :

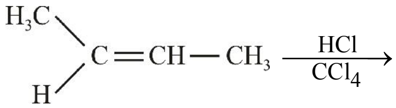
(a) 

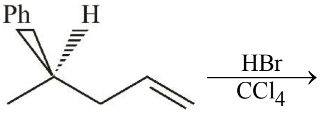
(b) 

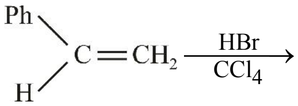
(c) 

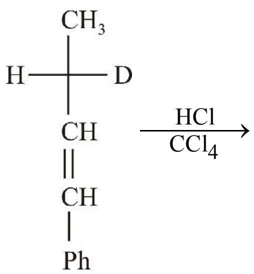
(d) 

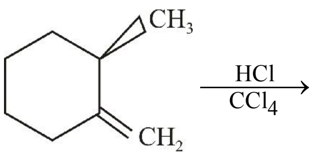
13. (I) How many of the following reactions, leads to the formation of diastereomers.

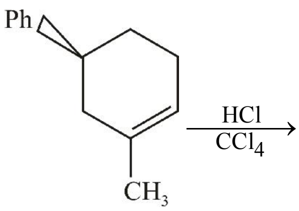
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

14. (S) When but –3-*en* -2- *ol* reacts with aq. HBr, the product formed is

(a) 3 - bromobut - 1- ene

(b) 1 - bromobut - 2- ene

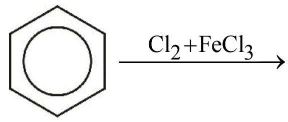
(c) A mixture of both a and b

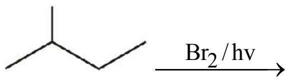
(d) 2 - bromobut - 2 - ene

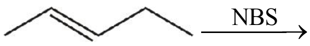
15. (X)

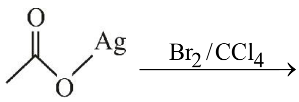
Column I	**Column II**
(A) $CH_3CH=CH_2 \xrightarrow{HBr,\ RCOOR}$	(p) Side chain substitution
(B) $CH_3CH=CH_2 \xrightarrow{HBr}$	(q) Anti-Markovnikov's addition
(C) $C_6H_5CH_3 \xrightarrow{NBS,\ h\nu}$	(r) Aromatic Substitution
(D) $C_6H_5CH_3 \xrightarrow{Br_2/FeBr_3}$	(s) Markovnikov's addition

16. (M) Which of the following reactions involve free radical as intermediate ?

(a) 

(b) 

(c) 

(d) 

17. (A) **Assertion :** Chlorination of allylic hydrogen is difficult than vinylic hydrogen.

Reason : Allyl radical is stabilised by resonance.

(a) A (b) B

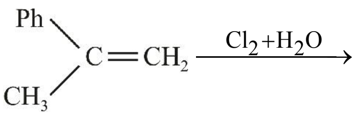
(c) D (d) D

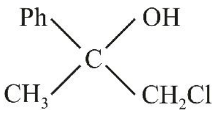
18. (A) **Assertion :** $CH_3-CH_2-Cl+NaI \xrightarrow{Acetone} CH_3-CH_2-I+NaCl\downarrow$

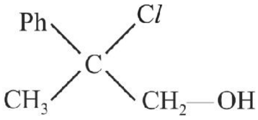
Reason : Acetone is polar protic solvent and solubility order of sodium halides decreases dramatically in order NaI > NaBr > NaCl. The last being virtually insoluble in this solvent and a 1° and 2° chloro alkane in acetone is completely driven to the side of Iodoalkane by the precipitation reaction.

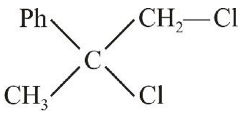
(a) A (b) B

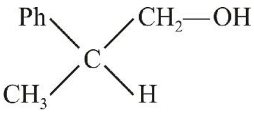
(c) C (d) D

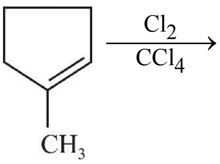
19. (S)  Major product :

(a) 

(b) 

(c) 

(d) 

20. (S) 

Stereochemistry of the product are :

(a) diastereomers (b) meso

(c) racemic mixture

(d) pure enantiomers

Reactions of Alkyl Halides : Nucleophilic Substitution

21. (S) An organic compound C_3H_7Br(A) with alcoholic KOH forms C_3H_6 which decolorises Br_2 in CCl_4 but does not gives a white precipitate with ammonical $AgNO_3$. (A) on reaction with KCN forms a product which on reduction with Na/C_2H_5OH produces n-butyl amine. Hence, the compound (A) is

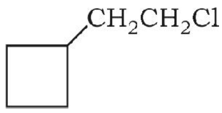
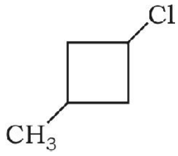
(a) n-propyl bromide

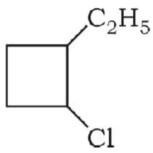
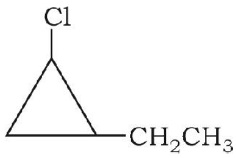
(b) isopropyl bromide

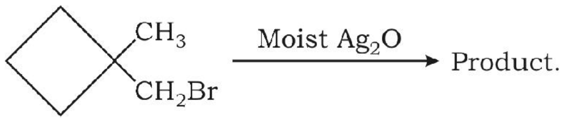
(c) 1, 1–dibromopropane

(d) none of these

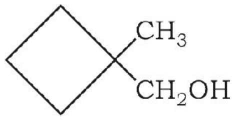
22. (S) A compound 'A' has molecular formula C_5H_9Cl. It does not react with bromine in CCl_4. On treatment with strong base, it produces single compound 'B' (C_5H_8) and reacts with Br_2 (aq). Ozonolysis of 'B' produces a compound $C_5H_8O_2$. Which of the following is structure of A ?

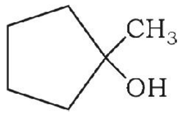
(a)  (b) 

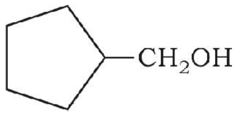
(c)  (d) 

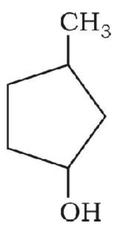
23. (M) 

Mark out the possible product.

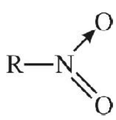
(a) 

(b) 

(c) 

(d) 

24. (M) Which of the following pair is correctly matched ?

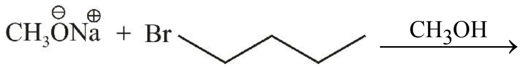
Reaction	Product
(a) RX + AgCN	RNC
(b) RX + KCN	RCN
(c) RX + KNO_2	
(d) RX + $AgNO_2$	R—O—N=O

25. (A) Assertion : Treatment of chloroethane with a saturated solution of AgCN gives ethyl isocyanide as the major product.

Reason : Cyanide (CN^-) is an ambident nucleophile.

(a) A (b) B
(c) C (d) D

26. (S) Choose the correct statement(s) about the following reaction.



I. The major product of the reaction is butyl-methyl ether.

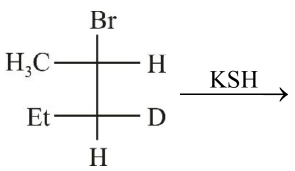
II. The major product of the reaction is 1-butene.

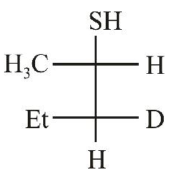
III. The major product is formed by S_{N^2} reaction mechanism.

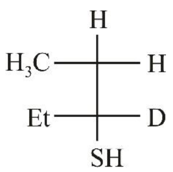
IV. The major product is followed by E2 reaction mechanism.

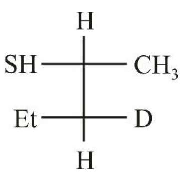
(a) I + III (b) II + IV
(c) I + IV (d) IV

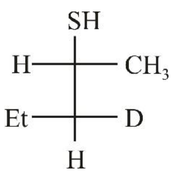
27. (S) The major substitution product of the following reaction is



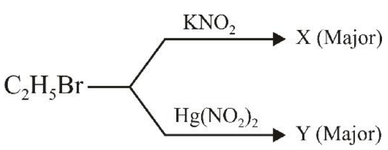
(a) 

(b) 

(c) 

(d) 

28. (S) Consider the following reaction



X and Y are respectively

(a) both nitroethane

(b) both ethyl nitrite (EtONO

(c) X = EtONO and Y = $EtNO_2$

(d) X = $EtNO_2$ and Y = EtONO

Reactions of Alkyl Halides : Metals

29. (S) Alkyl halides react with dialkyl copper reagents to give

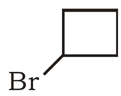
(a) Alkenes

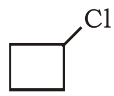
(b) Alkyl copper halides

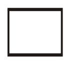
(c) Alkanes

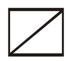
(d) Alkenyl halides

30. (S) What would be the product formed when 1-Bromo-3-chloro cyclobutane reacts with two equivalents of metallic sodium in ether

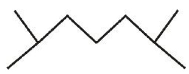
(a) 

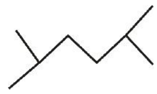
(b) 

(c) 

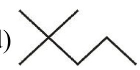
(d) 

31. (M) Which of the following alkanes cannot be synthesized by the Wurtz reaction in good yield ?

(a) 

(b) 

(c) 

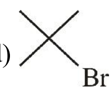
(d) 

32. (S) Which of the following alkyl halides is not suitable for Corey-House synthesis of alkanes ?

(a) CH_3I

(b) 

(c) 

(d) 

33. (X) Match the column :

Column-I Wurtz reaction	Column-II Number of dimerization product
(A) $CH_3-Cl \xrightarrow[\text{dry ether}]{Na}$	(p) 5
(B) $CH_3-Cl + CH_3-CH_2-Cl \xrightarrow[\text{dry ether}]{Na}$	(q) 6
(C) $CH_3-Cl + CH_3-CH_2-Cl + CH_3-CH_2-CH_2-Cl \xrightarrow[\text{dry ether}]{Na}$	(r) 3
(D) $H_2C=CH-CH=CH-CH_2-Cl + CH_3-CH_2-Cl \xrightarrow[\text{dry ether}]{Na}$	(s) 1

34. (A) **Assertion :** Wurtz's coupling reaction of alkyl halide is not suitable for an alkane containing an odd number of carbons in the parent chain.

Reason : Wurtz's coupling reaction involves heating of alkyl halides with sodium metal in ethereal solution.

(a) A (b) B

(c) C (d) D

35. (I) If an etheral solution containing 2–chloropentane and chloroethane is heated with sodium, how many different alkanes would result through Wurtz's reaction ?

Reactions of Alkyl Halides : Grigand Reagent

36. (S) $A \xrightarrow{H_3O^{\oplus}} CH_3-\underset{\large CH_3}{\underset{|}{C}}HCOOH$

$A \xrightarrow[H_3O^{\oplus}]{CH_3MgBr\ (2\ mol)} B$. 'B' is :

(a) $(CH_3)_2CHCOCH_3$

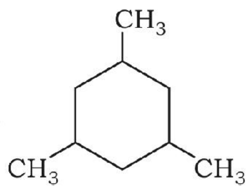
(b) $(CH_3)_2CH\underset{\large OH}{\underset{|}{C}}(CH_3)_2$

(c) $(CH_3)_2CHCOCH(CH_3)_2$

(d) $(CH_3)_2CH\underset{\large OH}{\underset{|}{C}}HCH(CH_3)_2$

37. (S) $CH_3CN \xrightarrow[\text{(ii) } H_3O^{\oplus}]{\text{(i) } CH_3MgBr\ (2\ mol)} A \xrightarrow{\text{conc. } H_2SO_4} B$,

A and B are

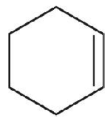
(a) CH_3COCH_3, 

(b) CH_3CH_2OH, $CH_2{=}CH_2$

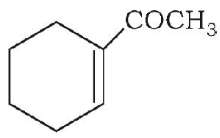
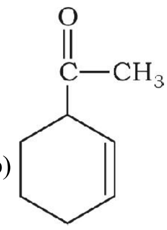
(c) $(CH_3)_2CHOH$, $CH_3CH{=}CH_2$

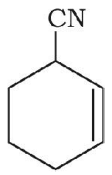
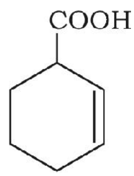
(d) $(CH_3)_3C-OH$, $(CH_3)_2C{=}CH_2$

38. (S) End product of the following sequence of reaction is



$\xrightarrow{NBS} \xrightarrow{Mg/ether} \xrightarrow[H_3O^{\oplus}]{CH_3CN}$

(a)  (b) 

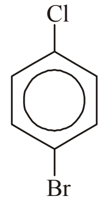
(c)  (d) 

39. (S) Which of these do not form Grignard reagent

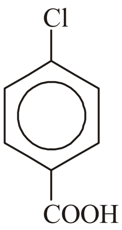
(a) CH_3F (b) CH_3Cl

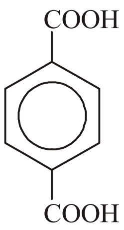
(c) CH_3Br (d) CH_3I

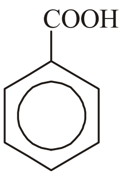
40. (S) Identify (Z) in the following reaction

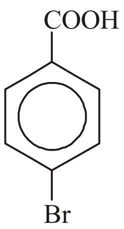


$+ \underset{1\text{ mole}}{Mg} \xrightarrow{THF} (X) \xrightarrow{CO_2} (Y) \xrightarrow{H^{\oplus}} (Z)$

(a) 

(b) 

(c) 

(d) 

41. (M) In the given reaction,

$(X) + R\,MgX\,(\text{excess}) \xrightarrow{H^+/H_2O} 3^\circ\text{ Alcohol}$

(X) may be :

(a) $R-\overset{O}{\overset{\|}{C}}-Cl$ (b) $R-\overset{O}{\overset{\|}{C}}-OC_2H_5$

(c) $R-\overset{O}{\overset{\|}{C}}-R$ (d) $R-\overset{O}{\overset{\|}{C}}-H$

42. (M) Which of the following combinations give

$C_6H_5-\underset{CH_2CH_3}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-OH$

(a) PhMgBr and $H_3C-\overset{O}{\overset{\|}{C}}-CH_2CH_3$

(b) CH_3MgBr and $Ph-\overset{O}{\overset{\|}{C}}-CH_2CH_3$

(c) CH_3MgBr and $C_6H_5-\overset{O}{\overset{\|}{C}}-OC_2H_5$

(d) CH_3CH_2MgBr and $Ph-\overset{O}{\overset{\|}{C}}-CH_3$

43. (M) $R-X+Mg \xrightarrow{\text{ether}} R-MgX \xrightarrow{CH_3OH} \text{n-butane}$

What can be R in the above reaction sequence ?

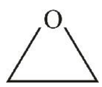
(a) n–propyl (b) n–butyl

(c) sec–butyl (d) Isopropyl

44. (I) $Cl-\overset{O}{\overset{\|}{C}}-OC_2H_5 \xrightarrow[NH_4Cl]{(1)\,'X'RMgX} 3^\circ$ Alcohol. Find out value of 'X'.

45. (M) Which of the following compounds give alcohol on reaction with RMgX ?

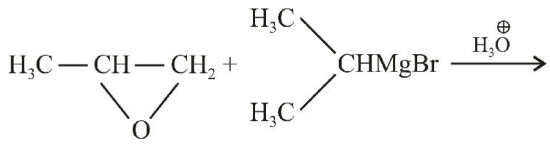
(a) O_2 (b) $R-\overset{O}{\overset{\|}{C}}-H$

(c) $R-O-H$ (d) 

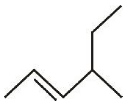
46. (S) $Cl-\overset{O}{\overset{\|}{C}}-OEt \xrightarrow[H_3^+O]{PhMgBr\,(\text{excess})}$ Product :

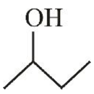
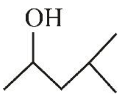
(a) $Ph-\underset{Ph}{\underset{|}{\overset{OH}{\overset{|}{C}}}}-Ph$ (b) $Ph-\overset{O}{\overset{\|}{C}}-Cl$

(c) $Ph-\overset{O}{\overset{\|}{C}}-OEt$ (d) $Ph-\overset{O}{\overset{\|}{C}}-Ph$

47. (S) 

(a) 

(b) 

(c)  (d) 

48. (X) Match the Column

Column-I	**Column-II**
(Reactant)	**(Reactant)**

(A) PhMgBr +

(A) $\xrightarrow[H^{\oplus}]{}$ 1° alcohol (p) $CH_3-\overset{\overset{\large O}{\|}}{C}-CH_2-\overset{\overset{\large O}{\|}}{C}-CH_3$

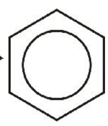
(B) PhMgBr +

(B) $\xrightarrow[H^{\oplus}]{}$ 2° alcohol (q) $CH_3-\overset{\overset{\large O}{\|}}{C}-CH_3$

(C) PhMgBr +

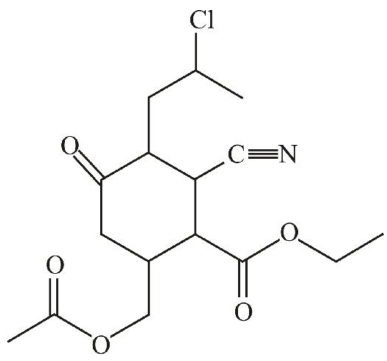
(C) $\xrightarrow[H^{\oplus}]{}$ 3° alcohol (r) $CH_3-\overset{\overset{\large O}{\|}}{C}-H$

(D) PhMgBr +

(D) $\xrightarrow[H^{\oplus}]{}$  (s) $H-\overset{\overset{\large O}{\|}}{C}-H$

Match the missing reagent A, B, C, D

49. (I) Total number of RMgX are consumed in the following reaction



Comprehension

Grignard reagent is usually prepared by

$R-X+Mg\xrightarrow{Et_2O}RMgX$

Grignard reagent

$Ar-X+Mg\xrightarrow{Et_2O}ArMgX$

Grignard reagent

Grignard reagent acts as a strong base. Grignard reagent carry out nucleophilic attact k in absence of acidic hydrogen. Grignard reagent form complex with its ether solvent. Complex formation with molecule of ether is an important factor in the formation and stability of Grignard reagent.

50. (C) What is the correct order of reactivity of halides with magnesium ?

(a) $R-Cl>R-Br>R-I$

(b) $R-Br>R-Cl>R-I$

(c) $R-I>R-Br>R-Cl$

(d) $R-I=R-Br=R-Cl$

51. (C) $Cl-\overset{\overset{\large O}{\|}}{C}-O-Et\xrightarrow[(2)NH_4Cl]{(1)xRMgX}$ 3° alcohol. Value of x is :

(a) 2 (b) 3

(c) 4 (d) 5

52. (C) $H-O-CH_2-CH_2-\overset{\overset{\large O}{||}}{C}-O-Et$

$$\xrightarrow[(2)H^+]{(1)x\,PhMgBr} HO-CH_2-CH_2-\underset{\underset{Ph}{|}}{\overset{\overset{OH}{|}}{C}}-Ph,$$

Value of x is :

(a) 2 (b) 3

(c) 4 (d) 5

Reactions of Aryl Halides

53. (S) Replacement of Cl of chlorobenzene to give phenol requires drastic conditions but chlorine of 2, 4-dinitrochlorobenzene is readily replaced because

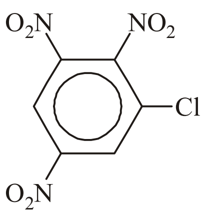
(a) NO_2 make ring electron rich at ortho and para

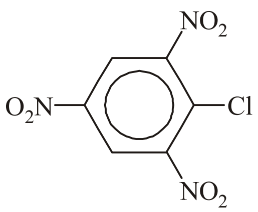
(b) NO_2 withdraws e^- from meta position

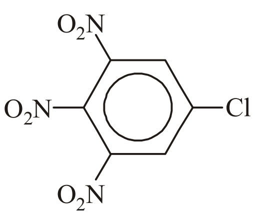
(c) denotes e^- at meta position

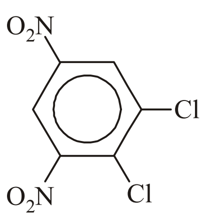
(d) NO_2 withdraws e^- from ortho/para positions

54. (S) Which chloroderivative of nitrobenzenes among the following would undergo hydrolysis, most readily with aqueous NaOH ?

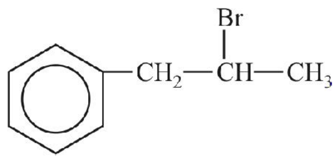
(a) 

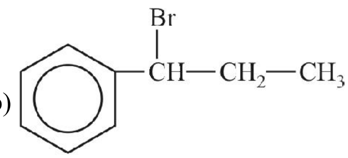
(b) 

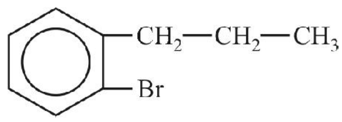
(c) 

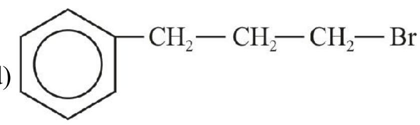
(d) 

55. (S) Propyl benzene reacts with bromine in presence of light or heat to give

(a) 

(b) 

(c) 

(d) 

56. (S) The reaction of 4-bromobenzyl chloride with NaCN in ethanol leads to

(a) 4-bromo-2-cyanobenzyl chloride

(b) 4-cyanobenzyl cyanide

(c) 4-cyanobenzyl chloride

(d) 4-bromobenzyl cyanide

57. (A) **Assertion :** The presence of nitro group facilitates nucleophilic substitution reaction in aryl halides.

Reason : The intermediate carbanion is stabilized due to the presence of the nitro-group.

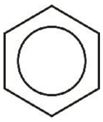
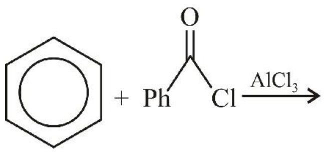
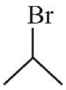
(a) A (b) B

(c) C (d) D

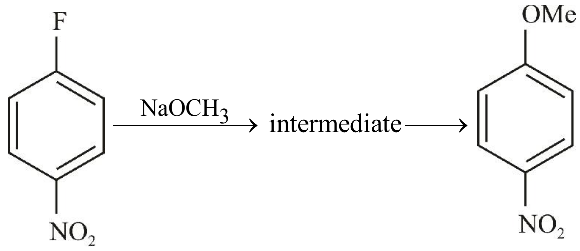
58. (X)

Column I	**Column II**
(A) Sandmeyer reaction	(p) C_6H_5COOAg, Br_2/CCl_4, heat
(B) Balz-Schiemann reaction	(q) $CH_3CH_2CH_2Br$, KI, acetone, heat
(C) Hunsdiecker reaction	(r) $C_6H_5N_2Cl$, HBF_4, heat
(D) Finkelstein reaction	(s) $C_6H_5N_2Cl$, CuCl/HCl, heat

59. (X) Match the column

Column - I	**Column - II**
(A)  $+ Ph-CH_2-Cl \xrightarrow{AlCl_3}$	(p) Nucleophilic substitution
(B) 	(q) Electrophilic substitution
(C)  $\xrightarrow{KOH(aq.)}$	(r) Cation intermediate
(D) $H_3C-\underset{CH_3}{\overset{CH_3}{C}}-CH_3 \xrightarrow{Br_2, hv}$	(s) Free radical substitution

60. (S) Which of the following is (are) true regarding intermediate in the addition-elimination mechanism of the reaction below ?



I. The intermediate is aromatic.

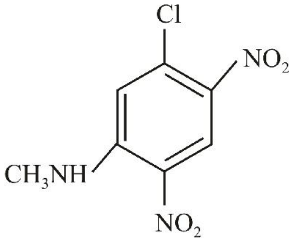
II. The intermediate is resonance stabilized anion.

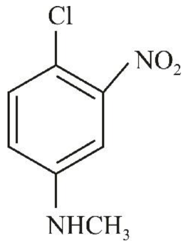
III. Electron withdrawing group on the benzene ring stabilize the intermediate

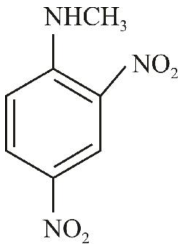
(a) I only (b) II only

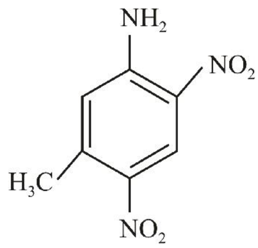
(c) I and III (d) II and III

61. (S) Which is the major product of the reaction of 2, 4–dinitrochlorobenzene with methylamine ?

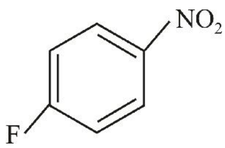
(a) 

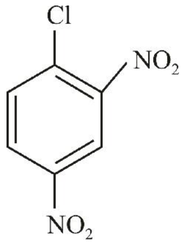
(b) 

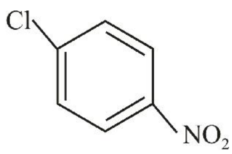
(c) 

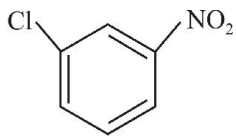
(d) 

62. (S) Arrange the following in increasing order of reactivity in aromatic nucleophilic substitution reaction.

(I) 

(II) 

(III) 

(IV) 

(a) I < IV < III < II (b) IV < III < I < II

(c) I < IV < II < III (d) IV < I < II < III

63. (M) p–nitrofluorobenzene is more reactive than its meta isomer in S_NAr reaction with CH_3ONa because

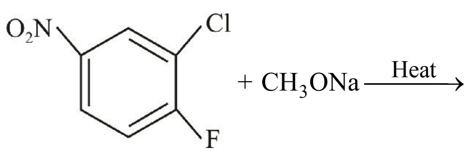
(a) nitro group from para position is in resonance with lone pair of fluorine.

(b) nitro group from para position stabilizes the benzene carbanion intermediate.

(c) nitro group from para position exert greater electronwithdrawing inductive effect than from meta position

(d) nitro group from para position gives less steric hindrance than from meta position to the attack of nucleophile.

64. (M) Consider the following S_NAr treaction



The correct statement is (are)

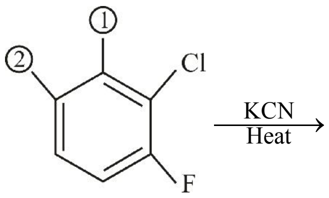
(a) Cl will be substituted because it is a better leaving group.

(b) F will be substituted because nitro from para position facilitate this.

(c) after substitution of one halogen, substitution of the second halogen would become further more difficult.

(d) first chlorine and then fluorine would be substituted subsequently.

65. (M) Consider the following chlorofluorobenzene undergoing nucleophilic aromatic (S_NAr) substitution reaction with KCN



The correct statement(s) is/are :

(a) substitution by a chlorine at '1' will promote substitution of fluorine by CN^-.

(b) substitution by a nitro group at '2' will promote substitution of fluorine by CN^-.

(c) substitution by a cyano group at '1' will promote substitution of chloro by CN^-.

(d) substitution by a methoxy group at either '1' or '2' will discourage the reaction as a whole.

66. (A) **Assertion :** p-nitrochlorobenzene is more reactive than m-nitrochlorobenzene towards aromatic nucleophilic substitution reaction.

Reason : Nitro group from para and meta positions has opposite effect in aromatic nucleophilic substitution reaction.

(a) A (b) B

(c) C (d) D

67. (A) **Assertion :** A dichlorobenzen is slightly more reactive than chlorobenzene towards an aromatic nucleophilic substitution reaction.

Reason : Chlorine has electron withdrawing effect on aromatic ring.

(a) A (b) B

(c) C (d) D

Polyhalogen Compounds

68. (S) Use of chlorofluoro carbons is not encouraged because

(a) They are harmful to the eyes of people that use it

(b) They damage the refrigerators and air conditioners

(c) They eat away the ozone in the atmosphere

(d) They destroy the oxygen layer

69. (S) The use of the product obtained as a result of reaction between acetone and chloroform is

(a) Hypnotic (b) Antiseptic

(c) Germicidal (d) Anaesthetic

70. (S) B.H.C. is used as

(a) Insecticide (b) Pesticide

(c) Herbicide (d) Weedicide

71. (S) In fire extinguisher, pyrene is

(a) CO_2 (b) CCl_4

(c) CS_2 (d) $CHCl_3$

72. (S) The commercial uses of DDT and benzene hexachloride are

(a) DDT is a herbicide, benzene hexachloride is a fungicide

(b) Both are insecticides

(c) Both are herbicides

(d) DDT is a fungicide and benzene hexachloride is a herbicide

73. (S) Which plastic is obtained from $CHCl_3$ as follows

$$CHCl_3 \xrightarrow[SbF_3]{HF} X \xrightarrow{800^oC} Y \xrightarrow{Polymerisation} \text{Plastic}$$

(a) Bakelite (b) Teflon

(c) Polythene (d) Perspex

74. (S) What is the reagent used for testing fluoride ion in water

(a) Alizarin-*S* (b) Quinalizarin

(c) Phenolphthalein (d) Benzene

75. (S) CCl_4 is a well known fire extinguisher. However after using it to extinguish fire, the room should be well ventilated. This is because :

(a) it is inflammable at higher temperatures

(b) it is toxic

(c) it produces phosgene by reaction with water vapours at higher temperatures

(d) it is corrosive

76. (S) When $CHCl_3$ is boiled with NaOH, It gives

(a) Formic acid (b) Trihydroxy methane

(c) Acetylene (d) Sodium formate

77. (A) **Assertion :** Chloral reacts with phenyl chloride to form DDT.

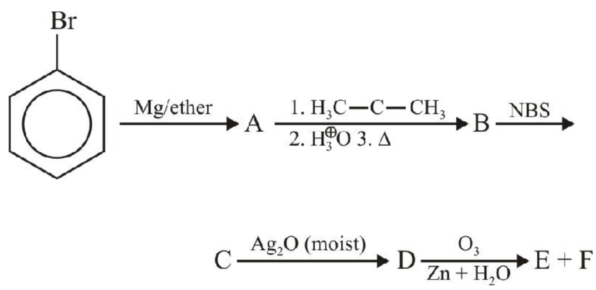
Reason : It is an electrophilic substitution reaction

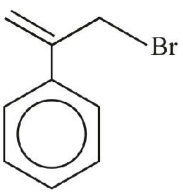
(a) A (b) B

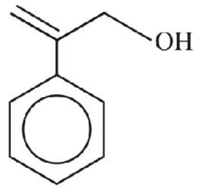
(c) C (d) D

Conversions/Roadmaps

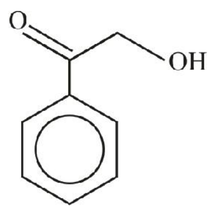
78. (M)



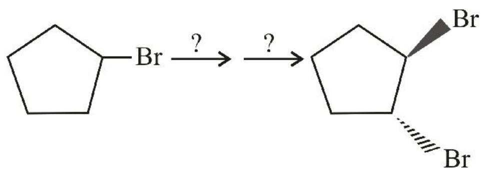
(a) Product C is 

(b) Product D is 

(c) Product E is $\mathrm{H_2C{=}O}$

(d) Product F is 

79. (S) Which of the following is missing reagent in the following reaction sequence ?



(a) $(CH_3)_3CO^-K^+$, Br_2 (b) $(CH_3)_3CO^-K^+$, HBr

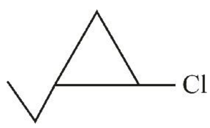
(c) $(CH_3)_3CO^-K^+$, Br_2/H_2O

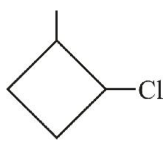
(d) H_2SO_4, Br_2

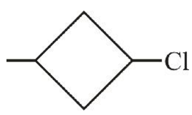
80. (S) $\underset{A}{C_5H_9Cl} \xrightarrow{\text{alc. KOH}} \underset{\text{(No isomers)}}{B(C_5H_8)} \xrightarrow[(CH_3)_2S]{O_3} C_5H_8O_2$

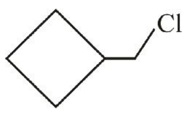
A $\downarrow Br_2, CCl_4$ Negative test; B $\downarrow Br_2, CCl_4$ Positive test

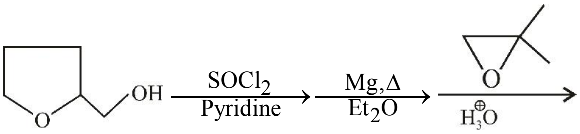
Which of the following is the structure of A ?

(a) 

(b) 

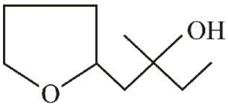
(c) 

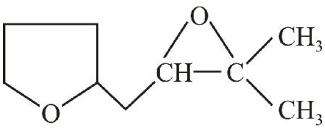
(d) 

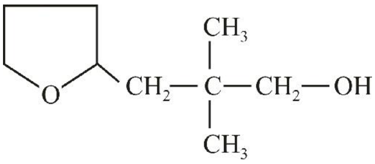
81. (S) 

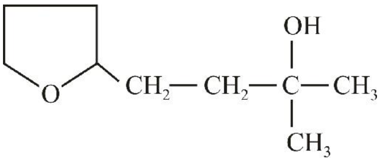
Product

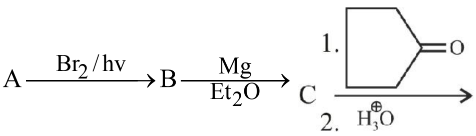
Product of reaction is :

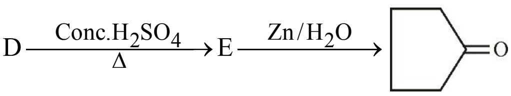
(a) 

(b) 

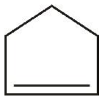
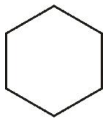
(c) 

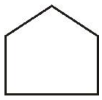
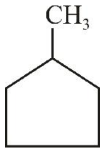
(d) 

82. (S) 

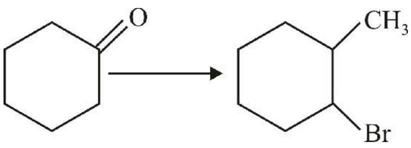


Find out the structure of 'A' :

(a)  (b) 

(c)  (d) 

83. (S) Which of the following sequence of reagent is best suited for the reaction shown below ?



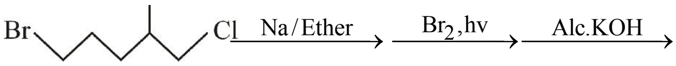
(a) (i) CH_3MgBr, H_3O^+ (ii) H^+/Δ (iii) HBr/H_2O_2

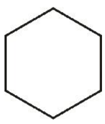
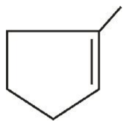
(b) (i) CH_3MgBr, H_3O^+ (ii) H^+/Δ (iii) HBr

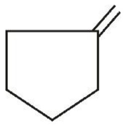
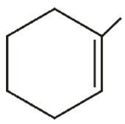
(c) (i) CH_3MgBr, H_3O^+ (ii) HBr

(d) (i) HBr/ROOR (ii) CH_3MgBr, H_3O^+

84. (S) The major product of the following reaction is



(a)  (b) 

(c)  (d) 

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

1. Which of the following has the highest nucleophilicity ? **(2000)**

(a) F^- (b) OH^-

(c) CH_3^- (d) NH_2^-

2. An S_N2 reaction of an asymmetric carbon of a compound always gives **(2001)**

(a) an enantiomer of the substrate

(b) a product with opposite optical rotation

(c) a mixture of diastereomers

(d) a single stereoisomer

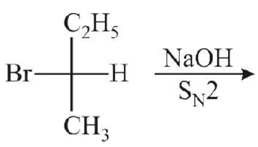
3. Among the following, the molecule with the highest dipole moment is **(2003)**

(a) CH_3Cl (b) CH_2Cl_2

(c) $CHCl_2$ (d) CCl_4

4. Write the structure of all the possible isomers of dichloroethene. Which of them will have zero dipole moment ? **(1985)**

5. Draw the stereochemical structures of the products in the following reactions



6. Write down the structures of the stereoisomers formed when cis-2-butene is reacted with bromine. **(1995)**

7. Chlorobenzene can be prepared by reacting aniline with **(1984)**

(a) hydrochloric acid

(b) cuprous chloride

(c) chlorine in presence of anhydrous aluminium chloride

(d) nitrous acid followed by heating with cuprous chloride

8. The compound which reacts fastest with Lucas reagent at room temperature is **(1981)**

(a) butan-2-ol (b) butan-1-ol

(c) 2-methyl propan-1-ol

(d) 2-methyl propan-2-ol

9. HBr reacts fastest with **(1986)**

(a) 2-methyl propan-2-ol (b) propan-1-ol

(c) propan-2-ol (d) 2-methyl propan-1-ol

10. The reaction conditions leading to the best yield of C_2H_5Cl are **(1986)**

(a) $C_2H_6 \text{ (excess)} + Cl_2 \xrightarrow{\text{UV light}}$

(b) $C_2H_6 + Cl_2 \text{ (excess)} \xrightarrow[\text{room temp.}]{\text{Dark}}$

(c) $C_2H_6 + Cl_2 \text{ (excess)} \xrightarrow{\text{UV light}}$

(d) $C_2H_6 + Cl_2 \xrightarrow{\text{UV light}}$

11. n-propyl bromide on treatment with ethanolic potassium hydroxide produces **(1987)**

(a) propane (b) propene

(c) propyne (d) propanol

12. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives **(1990)**

(a) o-cresol (b) p-cresol

(c) 2, 4-dihydroxy toluene

(d) benzoic acid

13. 1-chlorobutane on reaction with alcoholic potash gives **(1991)**

(a) but-1-ene (b) 1-butanol

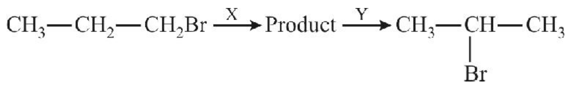
(c) but-2-ene (d) 2-butanol

14. $(CH_3)_3CMgCl$ on reaction with D_2O produces **(1997)**

(a) $(CH_3)_3CD$ (b) $(CH_3)_3COD$

(c) $(CD)_3CD$ (d) $(CD)_3COD$

15. Identify the set of reagents/reaction conditions X and Y in the following set of transformations **(2002)**



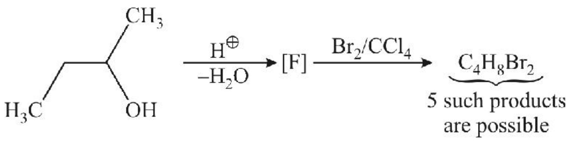
(a) X = dilute aqueous NaOH, 20°C, Y = HBr/acetic acid, 20°C

(b) X = concentrated alcoholic NaOH, 80°C, Y = HBr/acetic acid, 20°C

(c) X = dilute aqueous NaOH, 20°C, Y = $Br_2/CHCl_3$, 0°C

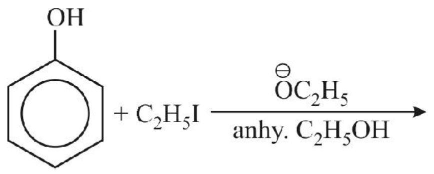
(d) X = concentrated alcoholic NaOH, 80°C, Y = $Br_2/CHCl_3$, 0°C

16. How many structures of F is possible ? **(2003)**



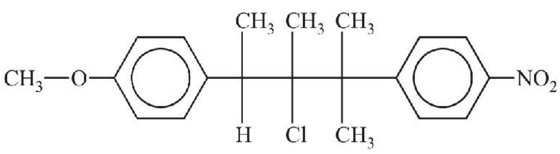
(a) 2 (b) 5
(c) 6 (d) 3

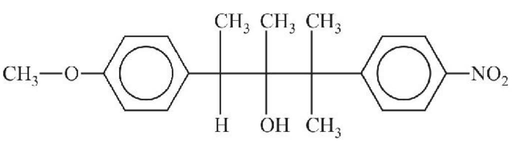
17. **(2003)**

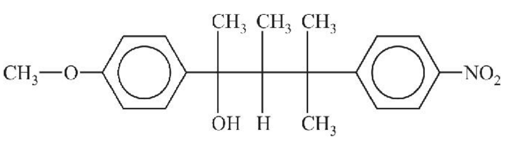


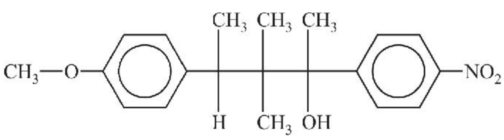
(a) $C_6H_5OC_2H_5$ (b) $C_2H_5OC_2H_5$
(c) $C_6H_5OC_6H_5$ (d) C_6H_5I

18. The following compound on hydrolysis in aqueous acetone will give **(2005)**



(K) 

(L) 

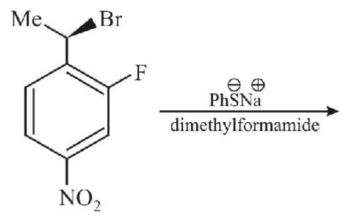
(M) 

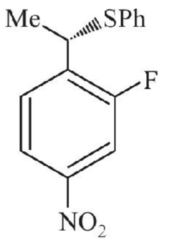
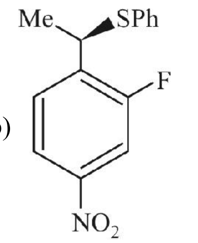
(a) mixture of (K) and (L)
(b) mixture of (K) and (M)
(c) only (M) (d) only (K)

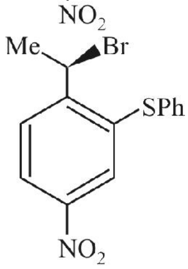
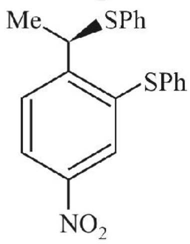
19. When phenyl magnesium bromide reacts with t-butanol, the product would be **(2005)**

(a) benzene (b) phenol
(c) t-butyl benzene (d) t-butyl phenyl ether

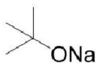
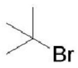
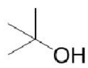
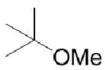
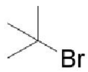
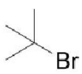
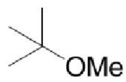
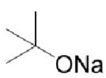
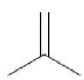
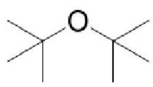
20. The major product of the following reaction is **(2008)**



(a)  (b) 

(c)  (d) 

21. List-I contains reactions and List-II contains major products. **(2018)**

List-I	List - II
(P)  +  →	1. 
(Q)  + HBr	2. 
(R)  + NaOMe	3. 
(S)  + MeBr	4. 
	5. 

Match each reaction in List-I with one or more products in List-II and choose the correct option.

(a) P - 1,5; Q - 2; R - 3; S - 4 (b) P - 1,4; Q - 2; R - 4; S - 3
(c) P - 1,4; Q - 1, 2; R - 3,4; S - 4 (d) P - 4,5; Q - 4; R - 4; S - 3,4

22. **Statement–I :** Aryl halides undergo nucleophilic substitution with ease. **(1991)**

Statement–II : The carbon-halogen bond in aryl halides has partial double bond character.

23. **Statement–I :** Benzonitrile is prepared by the reaction of chlorobenzene with potassium cyanide. **(1998)**

Statement–II : Cyanide (CN^-) is a strong nucleophile.

24. Aryl halide are less reactive towards nucleophilic substitution reaction as compared to alkyl halide due to **(1990)**

(a) the formation of less stable carbonium ion

(b) resonance stabilization

(c) longer carbon halogen bond

(d) the inductive effect

(e) sp^2-hybridised carbon attached to the halogen

25. The compounds used as refrigerant are **(1990)**

(a) NH_3 (b) CCl_4 (c) CF_4

(d) CF_2Cl_2 (e) CH_2F_2

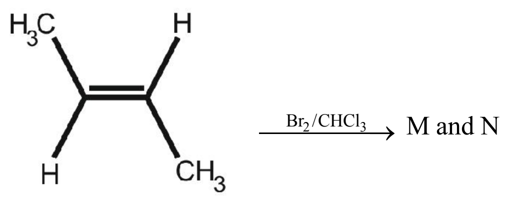
26. The products of reaction of alcoholic silver nitrite with ethyl bromide are **(1991)**

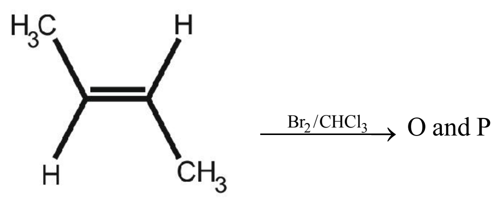
(a) ethane (b) ethene

(c) nitroethane (d) ethyl alcohol

(e) ethyl nitrite

27. The correct statements for the following addition reactions is (are) **(2017)**





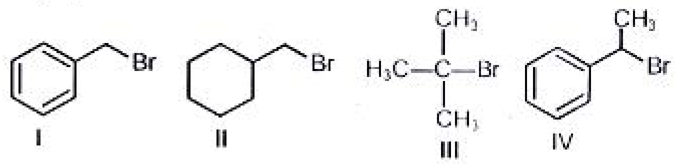
(a) (M and O) and (N and P) are two pairs of enantiomers

(b) Bromination proceeds through trans-addition in both the reactions

(c) O and P are identical molecules

(d) (M and O) and (N and P) two pairs of diastereomers

28. For the following compounds, the correct statements with respect to nucleophilic substitution reaction is (are)



(a) Compound IV undergoes inversion of configuration

(b) The order of reactivity for I, III and IV is : IV > I > III

(c) I and III follow S_N1 mechanism

(d) I and II follow S_N1 mechanism

29. The halogen which is most reactive in the halogenation of alkanes under sunlight is ... **(1981)**

30. The starting material for the manufacture of polyvinyl chloride is obtained by reacting HCl with ... **(1983)**

31. Carbon tetrachloride burns in air when lighted to give phosgene. **(1983)**

32. Carbon tetrachloride is inflammable. **(1985)**

33. The reaction of vinyl chloride with hydrogen iodide to give 1-chloro-1-iodoethane is an example of anti-Markownikoff's rule. **(1989)**

34. Photobromination of 2-methyl propane gives a mixture of 1-bromo-2-methyl propane and 2-bromo-2-methyl propane in the ratio 9 : 1. **(1993)**

35. Show by chemical equations only, how would you prepare the following from the indicated starting materials. Specify the reagents in each step of the synthesis. **(1979)**

(i) Chloroform from carbon disulphide.

(ii) Hexachloroethane (C_2Cl_6) from calcium carbide.

36. Chloroform is stored in dark coloured bottles. Explain in not more than two sentences. **(1980)**

37. Write down the reaction involved in the preparation of following using the reagents indicated against in parenthesis. **(1984)**

"Ethyl benzene from benzene."

[C_2H_5OH, PCl_5, anhyd. $AlCl_3$]

38. Arrange the following in order of their **(1992)**

(i) Increasing basicity

H_2O, OH^-, CH_3OH, CH_3O^-

(ii) Increasing reactivity in nucleophilic substitution reactions

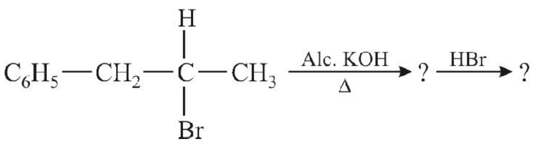
CH_3F, CH_3I, CH_3Br, CH_3Cl

39. Write the structure of the major organic product expected from each of the following reactions **(1992)**

(i) $(CH_3)_2\underset{\underset{Cl}{|}}{C}—CH_2—CH_3 \xrightarrow{\text{Alc. KOH}}$

(ii) $CH_3CH_2CHCl_2 \xrightarrow[\text{alkali}]{\text{Boil}}$

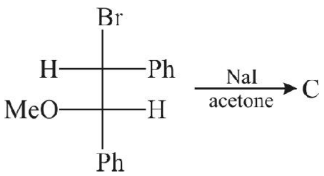
40. Identify the major product in the following reaction **(1993)**



41. Aryl halides are less reactive than alkyl halides towards nucleophilic reagents. Give reason. **(1994)**

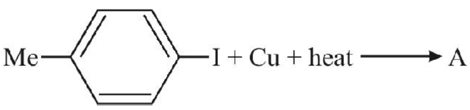
42. Optically active 2-iodobutane on treatment with NaI in acetone gives a product which does not show optical activity. Explain briefly. **(1995)**

43. Predict the structure of the product in the following reaction **(1996)**



44. How will you prepare m-bromoiodobenzene from benzene (in not more than 5-7 steps) ? **(1996)**

45. Complete the following, giving the structures of the principal organic products **(1997)**



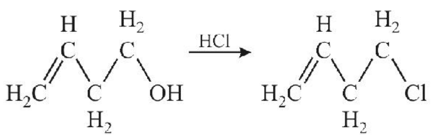
46. Each of the following reactions gives two products. Write the structures of the products **(1998)**

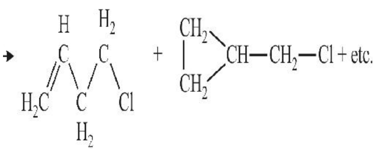
(i) $C_6H_5CH_2CHClC_6H_5 \xrightarrow{\text{Alc. KOH, heat}}$

(ii) $(CH_3)_2CH—OCH_3 \xrightarrow[\text{heat}]{\text{HI (excess)}}$

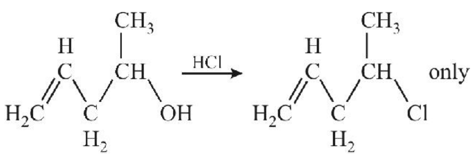
47. Explain briefly the formation of the products giving the structures of the intermediates. **(1999)**

(a) (i)

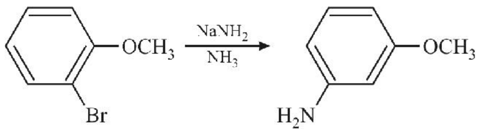




(ii)

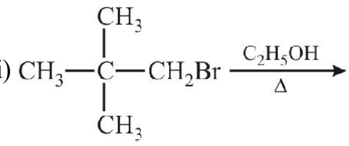


(b)

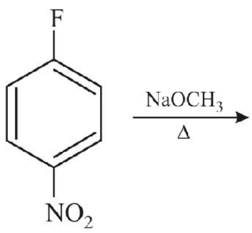


48. What would be the major product in each of the following reactions ? **(2000)**

(i)



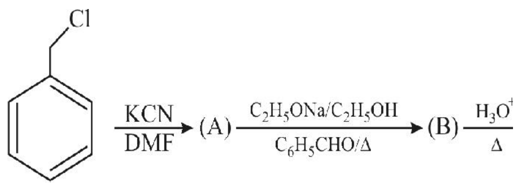
(ii)



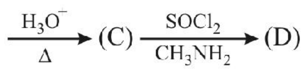
49. How would you synthesize 4-methoxyphenol from bormobenzene in NOT more than five steps ? State clearly the reagents used in each step and show the structures of the intermediate compounds in your synthetic scheme. **(2001)**

50. Cyclobutylbromide on treatment with magnesium in dry ether forms an organometallic compound (A). The organometallic reacts with ethanal to give an alcohol (B) after mild acidification. Prolonged treatment of alcohol (B) with an equivalent amount of HBr gives 1-bromo-1-methylcyclopentane (C). Write the structures of (A), (B) and explain how (C) is obtained from (B). **(2001)**

51.



(2004)



Identify A to D.

Answer Key

EXERCISE - 1 : (Basic Objective Questions)

1. (b) **2.** (d) **3.** (a) **4.** (b) **5.** (b) **6.** (b) **7.** (a) **8.** (d) **9.** (d) **10.** (a)
11. (a) **12.** (a) **13.** (b) **14.** (b) **15.** (b) **16.** (a) **17.** (b) **18.** (c) **19.** (d) **20.** (c)
21. (d) **22.** (b) **23.** (a) **24.** (bd) **25.** (d) **26.** (a) **27.** (c) **28.** (b) **29.** (a) **30.** (d)
31. (d) **32.** (d) **33.** (a) **34.** (c) **35.** (b) **36.** (a) **37.** (d) **38.** (c) **39.** (ad) **40.** (c)
41. (c) **42.** (b) **43.** (a) **44.** (d) **45.** (a) **46.** (b) **47.** (a) **48.** (d) **49.** (b) **50.** (a)
51. (c) **52.** (c) **53.** (c) **54.** (b) **55.** (d) **56.** (b) **57.** (d) **58.** (a) **59.** (d) **60.** (d)
61. (c) **62.** (a) **63.** (c) **64.** (a) **65.** (b) **66.** (a) **67.** (a) **68.** (b) **69.** (b) **70.** (b)
71. (c) **72.** (a) **73.** (b) **74.** (cd) **75.** (a) **76.** (b) **77.** (b) **78.** (b) **79.** (c) **80.** (c)
81. (a) **82.** (b) **83.** (c) **84.** (a) **85.** (b)

EXERCISE - 2 : (Previous Year JEE Mains Questions)

1. (d) **2.** (b) **3.** (c) **4.** (c) **5.** (c) **6.** (c) **7.** (b) **8.** (a) **9.** (b) **10.** (d)
11. (d) **12.** (a) **13.** (d) **14.** (d) **15.** (c) **16.** (a) **17.** (b) **18.** (a) **19.** (b) **20.** (b)
21. (a) **22.** (a) **23.** (b) **24.** (d)

JEE Mains Online

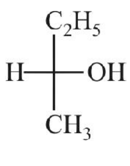
1. (a) **2.** (b) **3.** (c) **4.** (d) **5.** (c) **6.** (b) **7.** (b) **8.** (d) **9.** (a) **10.** (b)
11. (d) **12.** (b) **13.** (d) **14.** (b) **15.** (b) **16.** (c)

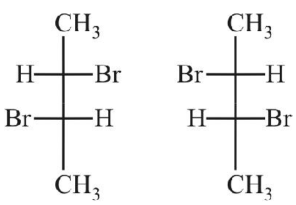
EXERCISE - 3 : (Advanced Objective Questions)

1. (a) **2.** (b) **3.** (a) **4.** (b) **5.** (c) **6.** (c) **7.** (c) **8.** (c) **9.** (b) **10.** (a)
11. (b) **12.** (d) **13.** (0004) **14.** (c) **15.** (A – q; B – s; C – p, D – r) **16.** (bcd) **17.** (d)
18. (c) **19.** (a) **20.** (c) **21.** (a) **22.** (b) **23.** (ab) **24.** (ab) **25.** (b) **26.** (a) **27.** (d)
28. (c) **29.** (c) **30.** (d) **31.** (a, c, d) **32.** (d) **33.** (A – s; B – r; C – p; D – q) **34.** (b) **35.** (0006)
36. (b) **37.** (d) **38.** (b) **39.** (a) **40.** (a) **41.** (a, b, c) **42.** (a, b, d) **43.** (b, c)
44. (0003) **45.** (a, b, d) **46.** (a) **47.** (d) **48.** (A – s; B – r; C – q; D – p) **49.** (0007) **50.** (c)
51. (b) **52.** (b) **53.** (d) **54.** (b) **55.** (b) **56.** (d) **57.** (a) **58.** (A – s; B – r; C – p; D – q)
59. (A – q, r; B – q, r; C – p; D – s) **60.** (d) **61.** (c) **62.** (b) **63.** (bd) **64.** (bc) **65.** (bcd) **66.** (c)
67. (a) **68.** (c) **69.** (a) **70.** (a) **71.** (b) **72.** (b) **73.** (b) **74.** (a) **75.** (c) **76.** (d)
77. (a) **78.** (abcd) **79.** (a) **80.** (c) **81.** (d) **82.** (c) **83.** (a) **84.** (b)

EXERCISE - 4 : (Previous Year IIT Questions)

1. (c) **2.** (d) **3.** (a) **4.** cis isomer & trans isomer. Trans isomer has zero dipole moment.

5. 

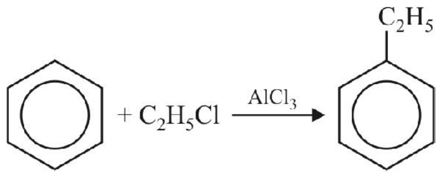
6. 

7. (d) **8.** (d) **9.** (a)

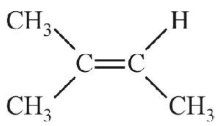
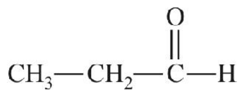
10. (a) **11.** (b) **12.** (d) **13.** (a) **14.** (a) **15.** (b) **16.** (d) **17.** (b)
18. (a) **19.** (a) **20.** (a) **21.** (b) **22.** (d) **23.** (d) **24.** (b, e) **25.** (a, d)

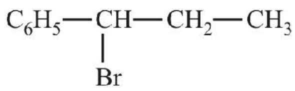
26. (c, e) **27.** (a,b,d) **28.** (a,d) **29.** fluorine **30.** C_2H_2 **31.** F **32.** F **33.** F

34. F **36.** chloroform reacts with O_2.

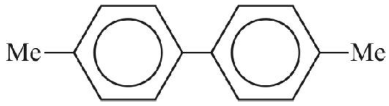
37. 

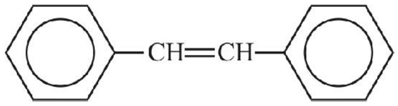
38. (i) $CH_3O^{\ominus} > OH^{\ominus} > CH_3OH > H_2O$; (ii) $CH_3F < CH_3Cl < CH_3Br < CH_3I$

39. (i)  (ii) 

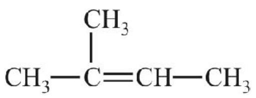
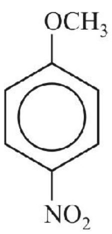
40. 

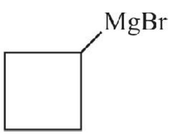
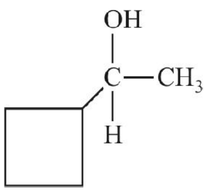
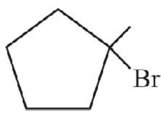
42. Formation of a racemic mixture.

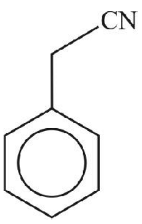
45. 

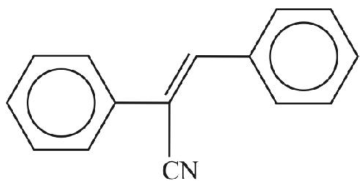
46. (i) 

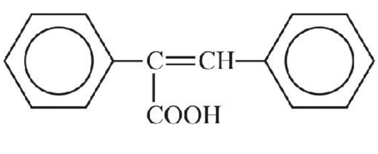
(ii) $(CH_3)_2CH-OH + CH_3I$

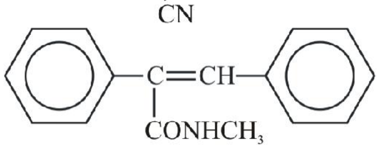
48. (i)  (ii) 

50. A :  B :  C : 

51. A : 

B : 

C : 

D : 

Dream on !!